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LTM

User Manual



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Chapter 1
About the Kit

The TRDB_LTM (LTM) Kit provides everything you need to develop applications using a digital touch panel on an Altera DE4/DE2-115/DE2/DE1 board. The kit contains complete reference designs and source code for implementing a photo viewer demonstration and a color pattern generator using the LTM kit and an Altera DE2-115/DE2/DE1 board. This chapter provides users key information about the kit.

1.1 Kit Contents

Figure 1.1 shows the picture of the TRDB_LTM package. The package includes:

1. The Terasic LCD Touch Panel Module (LTM) board
2. A 40-pin IDE cable
3. Complete reference design with source code

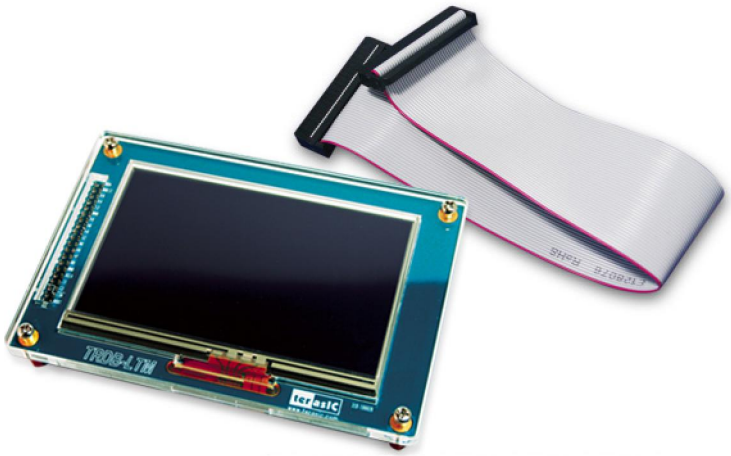


Figure 1.1. The TRDB_LTM Package

1.2 Connecting LTM to Altera DE Series Board

Please follow the two steps below to connect LTM to the Altera DE2-115/DE2/DE1 board:

1. Connect the IDE cable to the back of the LTM board, as shown in Figure 1.2

2. Connect the other end of the IDE cable to your DE4/DE2-115/DE2/DE1 board's innermost expansion port (JP1) as shown in Figure 1.3 and Figure 1.6

Figure 1.2 Connect the IDE cable to the TRDB_LTM board

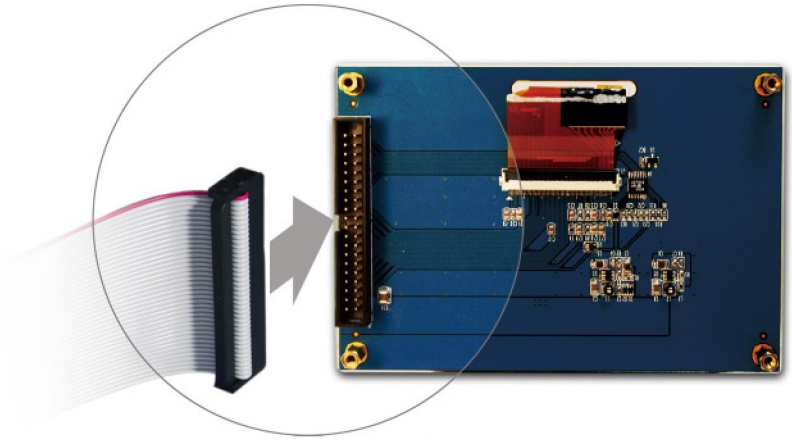


Figure 1.2 Connect the IDE cable to the TRDB_LTM board

Figure 1.3 Connect the other end of IDE cable to the DE2 board's expansion port (innermost expansion port)

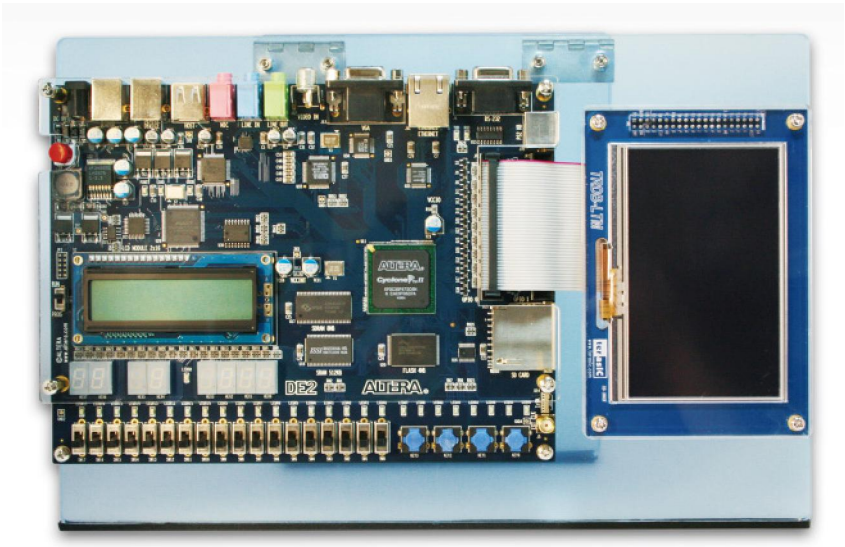


Figure 1.3 Connect the other end of IDE cable to the DE2 board's expansion port (innermost expansion port)

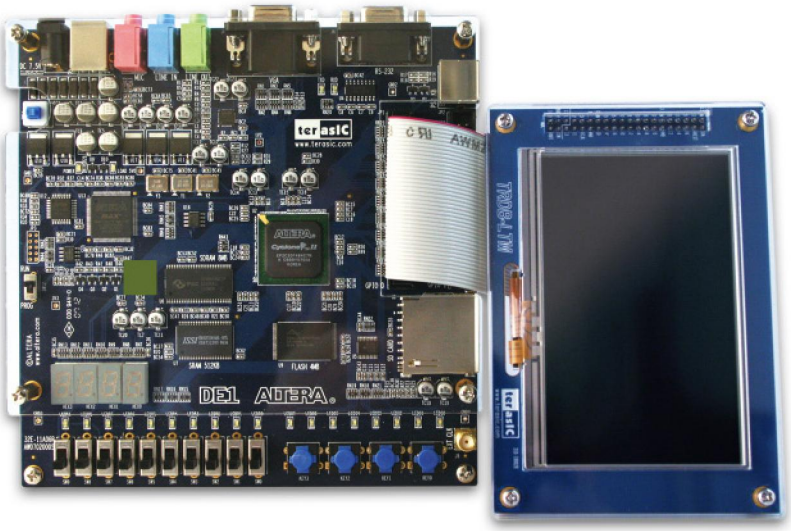


Figure 1.4 Connect the other end of IDE cable to the DE1 board's expansion port (innermost expansion port)

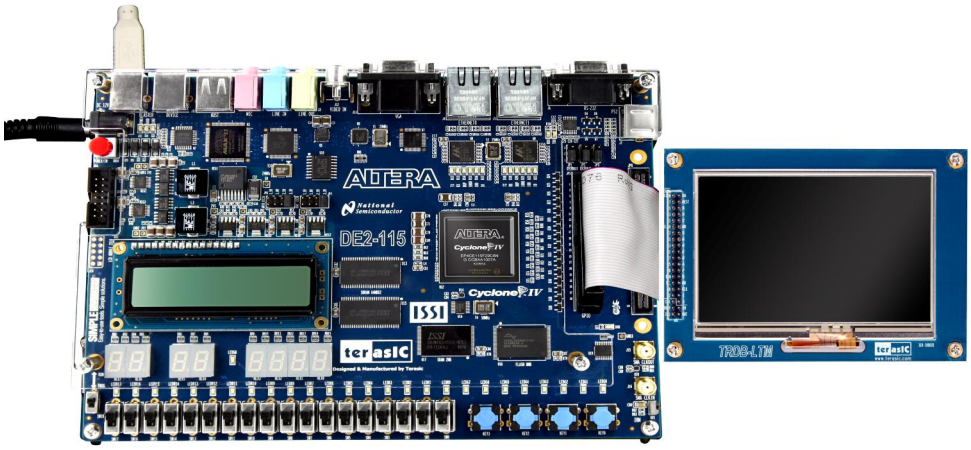


Figure 1.5 Connect the other end of IDE cable to the DE2-115 board's GPIO

Figure 1.6 Connect the other end of IDE cable to the DE4 board's expansion port (innermost expansion port)

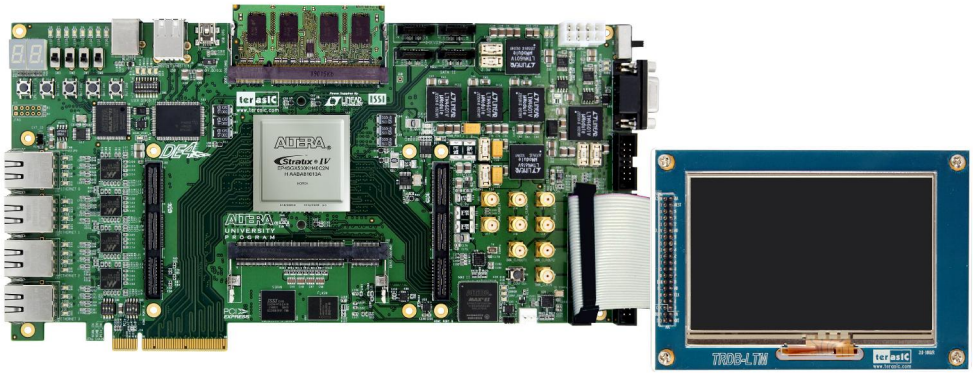


Figure 1.6 Connect the other end of IDE cable to the DE4 board's expansion port (innermost expansion port)

1.3 Getting Help

Here are some places to get help if you encounter any problem:

Email to support@terasic.com

Taiwan & China: +886-3-550-8800

Korea : +82-2-512-7661

Japan: +81-428-77-7000

Chapter 2
Architecture of the LTM

This chapter will illustrate the architecture of the LTM including device features and block diagram.

2.1 Features

The feature set of the LTM is listed below:

1. Equipped with Toppoly TD043MTEA1 active matrix color TFT LCD module.
2. Support 24-bit parallel RGB interface.
3. 3-wire register control for display and function selection.
4. Built-in contrast, brightness, and gamma modulation.
5. Converting the X/Y coordination of the touch point to its corresponding digital data via the Analog Devices AD7843 AD converter.
6. The general specifications of the LTM are listed below:

Item	Description	Unit
Display Size (Diagonal)	4.3	Inch
Aspect ratio	15:9	-
Display Type	Transmissive	-
Active Area (HxV)	93.6 x 56.16	mm
Number of Dots (HxV)	800 x RGB x480	dot
Dot Pitch (HxV)	0.039 x 0.117	mm
Color Arrangement	Stripe	-
Color Numbers	16Million	-

2.2 Block Diagram of the LTM

The block diagram of the LTM is listed below:

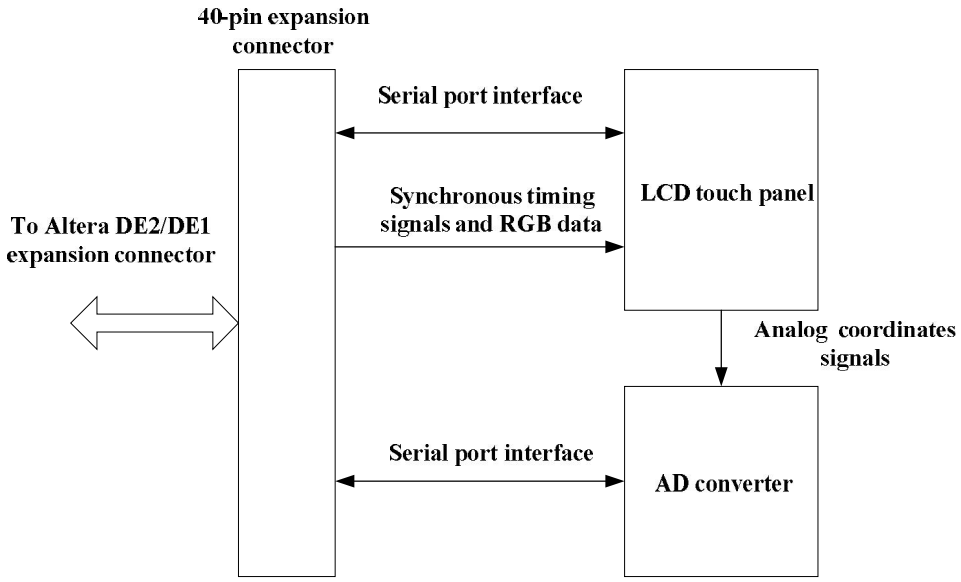


Figure 2.1 The block diagram of the LTM.

The LTM consists of three major components: LCD touch panel module, AD converter, and 40-pin expansion header. All of the interfaces on the LTM are connected to Altera DE4/DE2-115/DE2/DE1 board via the 40-pin expansion connector. The LCD and touch panel module will take the control signals provided directly from FPGA as input and display images on the LCD panel. Finally, the AD converter will convert the coordinates of the touch point to its corresponding digital data and output to the FPGA via the expansion header.

2.3 Pin Description of the 40-pin Interface of LTM

The pin description of the 40-pin connector follows:

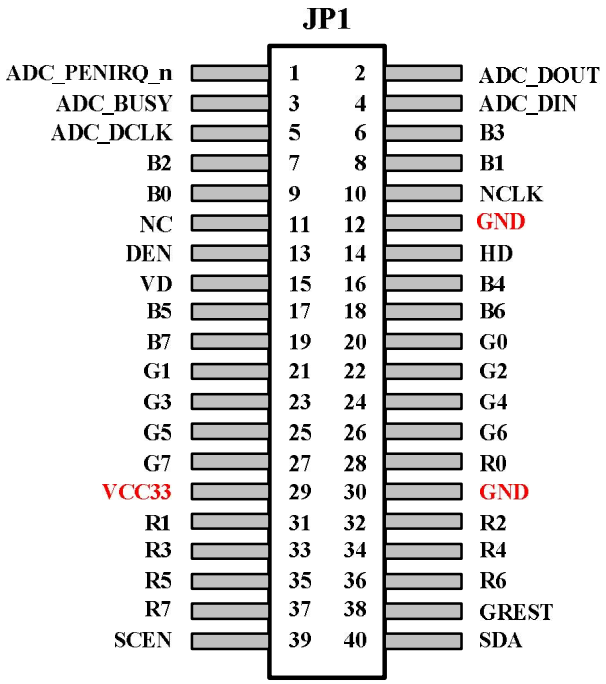


Figure 2.2 The pin-out of the 40-pin connector.

Pin Numbers	Name	Direction	Description
1	ADC_PENIRQ_n	output	ADC pen Interrupt
2	ADC_DOUT	output	ADC serial interface data out
3	ADC_BUSY	output	ADC serial interface busy
4	ADC_DIN	input	ADC serial interface data in
5	ADC_DCLK	input	ADC/LCD serial interface clock
6	B3	Input	LCD blue data bus bit 3
7	B2	Input	LCD blue data bus bit 2
8	B1	Input	LCD blue data bus bit 1
9	B0	Input	LCD blue data bus bit 0
10	NCLK	Input	LCD clock signal
11	NC	N/A	N/A
12	GND	N/A	Ground
13	DEN	Input	LCD RGB data enable
14	HD	Input	LCD Horizontal sync input
15	VD	Input	LCD Vertical sync input
16	B4	Input	LCD blue data bus bit 4
17	B5	Input	LCD blue data bus bit 5
18	B6	Input	LCD blue data bus bit 6
19	B7	Input	LCD blue data bus bit 7
20	G0	Input	LCD green data bus bit 0
21	G1	Input	LCD green data bus bit 1
22	G2	Input	LCD green data bus bit 2
23	G3	Input	LCD green data bus bit 3
24	G4	Input	LCD green data bus bit 4
25	G5	Input	LCD green data bus bit 5
26	G6	Input	LCD green data bus bit 6
27	G7	Input	LCD green data bus bit 7

28	R0	Input	LCD red data bus bit 0
29	VCC33	N/A	Power 3.3V
30	GND	N/A	Ground
31	R1	Input	LCD red data bus bit 1
32	R2	Input	LCD red data bus bit 2
33	R3	Input	LCD red data bus bit 3
34	R4	Input	LCD red data bus bit 4
35	R5	Input	LCD red data bus bit 5
36	R6	Input	LCD red data bus bit 6
37	R7	Input	LCD red data bus bit 7
38	GREST	Input	Global reset, low active
39	SCEN	Input	LCD 3-wire serial interface enable/ADC chip enable
40	SDA	Input/Output	LCD 3-wire serial interface data

Table 2.1 The pin description of the 40-pin connector.

Chapter 3
Using the LTM

This chapter illustrates how to use the LTM including how to control the serial port interface of the LCD driver IC and AD converter. Also, the timing requirement of the synchronous signal and RGB data which are outputted to the LCD panel module will be described.

3.1 The Serial Port Interface of the LCD Driver IC

The LCD and touch panel module on the LTM is equipped with a LCD driver IC to support three display resolutions and with functions of source driver, serial port interface, timing controller, and power supply circuits. To control these functions, users can use FPGA to configure the registers in the LCD driver IC via serial port interface.

Also, there is an analog to digital converter (ADC) on the LTM to convert the analog X/Y coordinates of the touch point to digital data and output to FPGA through the serial port interface of the ADC. Both LCD driver IC and ADC serial port interfaces are connected to the FPGA via the 40-pin expansion header and IDE cable.

Because of the limited number of I/O on the expansion header, the serial interfaces of the LCD driver IC and ADC need to share the same clock (ADC_DCLK) and chip enable (SCEN) signal I/O on the expansion header. To avoid both the serial port interfaces may interfere with each other when sharing the same clock and chip enable signals, the chip enable signal (CS), which is inputted into the ADC will come up with a logic inverter as shown in Figure 3.1.
Users need to pay attention controlling the shared signals when designing the serial port interface controller. The detailed register maps of the LCD driver IC are listed in appendix chapter. The specifications of the serial port interface of the LCD driver IC are described below.

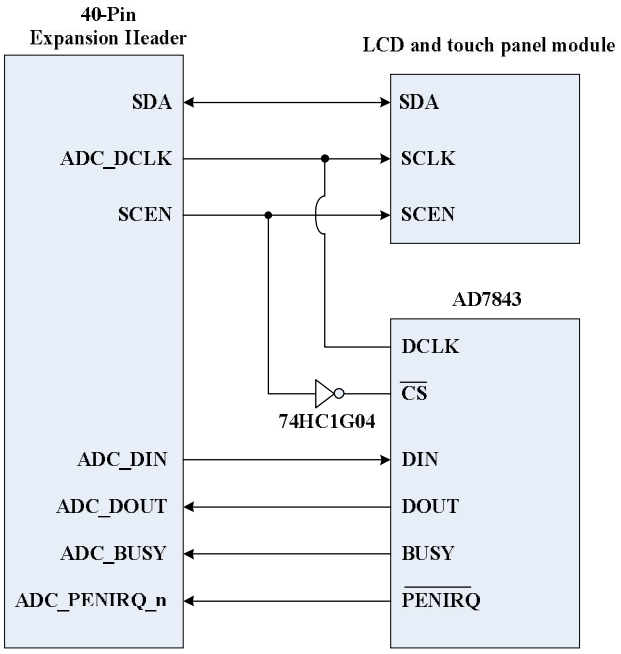


Figure 3.1 The serial interface of the LCD touch panel module and AD7843

The LCD driver IC supports a clock synchronous serial interface as the interface to a FPGA to enable instruction setting. Please notice that in addition to the serial port interface signals, NCLK input should also be provided while setting the registers. Figure 3.2 and Table 3.1 show the frame format and timing diagram of the serial port interface. The LCD driver IC recognizes the start of data transfer on the falling edge of SCEN input and starts data transfer. When setting instruction, the TPG110 inputs the setting values via SDA on the rising edge of input SCL.

The first 6 bits (A5 ~ A0) specify the address of the register. The next bit means Read/Write command. "0" is write command. "1" is read command. Then, the next cycle is turn-round cycle. Finally, the last 8 bits are for Data setting (D7 ~ D0). The address and data are transferred from the MSB to LSB sequentially. The data is written to the register of assigned address when "End of transfer" is detected after the 16th SCL rising cycles. Data is not accepted if there are less or more than 16 cycles for one transaction

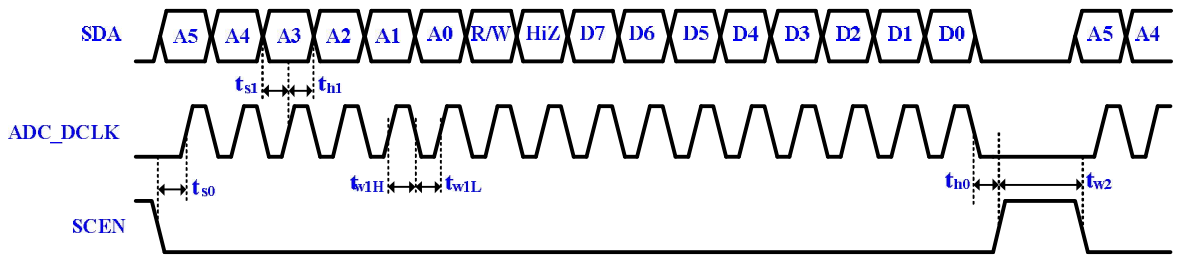


Figure 3.2 The frame format and timing diagram of the serial port interface

Item	Symbol	Conditions	Min.	Max.	Unit
SDA Setup Time	ts0	SCEN to SCL	150		ns
	ts1	SDA to SCL	150		ns
SDA Hold Time	th0	SCEN to SCL	150		ns
	th1	SDA to SCL	150		ns
Pulse Width	tw1L	SCL pulse width	160		ns
	tw1H	SCL pulse width	160		ns
	tw2	SCEN pulse width	1.0		ns
Clock duty			40	60	%

Table 3.1 The timing parameters of the serial port interface

3.2 Input timing of the LCD panel display function

This section will describe the timing specification of the LCD synchronous signals and RGB data.

To determine the sequencing and the timing of the image signals displayed on the LCD panel, the corresponding synchronous signals from FPGA to the LCD panel should follow the timing specification.

Figure 3.3 illustrates the basic timing requirements for each row (horizontal) that is displayed on the LCD panel. An active-low pulse of specific duration (time t_{hpw} in the figure) is applied to the horizontal synchronization (HD) input of the LCD panel, which signifies the end of one row of data and the start of the next. The data (RGB) inputs on the LCD panel are not valid for a time period called the hsync back porch (t_{hbp}) after the hsync pulse occurs, which is followed by the display area (t_{hd}). During the data display area the RGB data drives each pixel in turn across the row being displayed. Also, during the period of the data display area, the data enable signal (DEN) must be driven to logic high. Finally, there is a time period called the hsync front porch (t_{hfp}) where the RGB signals are not valid again before the next hsync pulse can occur.

The timing of the vertical synchronization (VD) is the same as shown in Figure 3.4, except that a vsync pulse signifies the end of one frame and the start of the next, and the data refers to the set of rows in the frame (horizontal timing). Table 3.2 and 3.3 show for different resolutions, the durations of time periods t_{hpw}, t_{hbp} , t_{hd} , and t_{hfp} for both horizontal and vertical timing. Finally, the timing specification of the synchronous signals is shown in the Table 3.4.

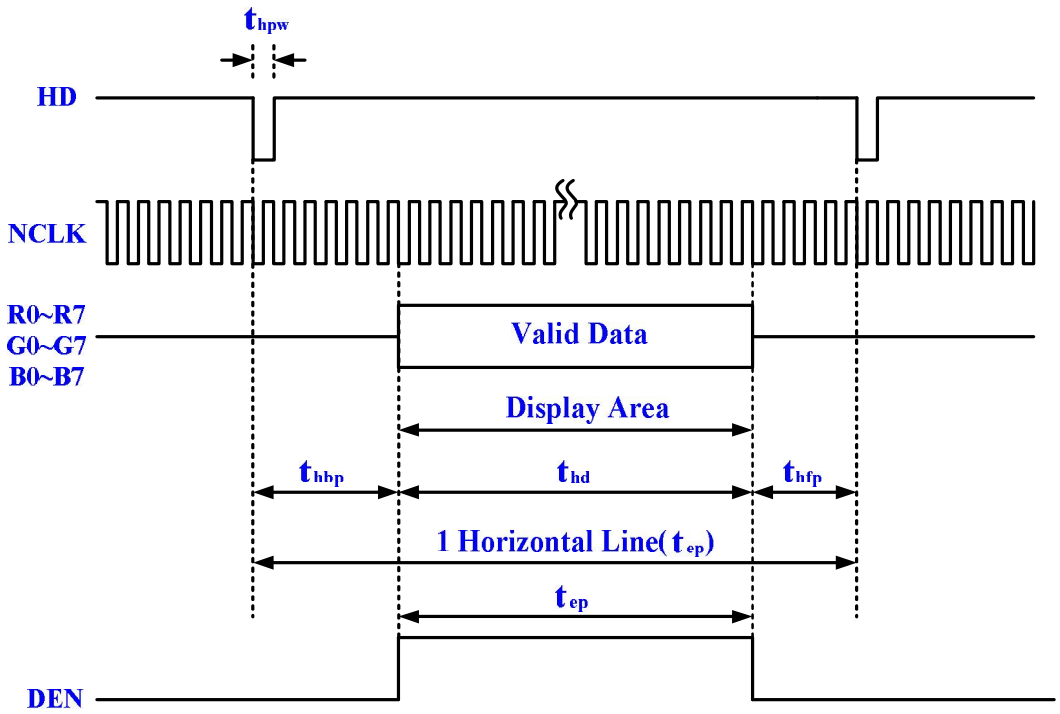


Figure 3.3 LCD horizontal timing specification

Parameter		Symbol	Panel Resolution			Unit
			800xRGBx480	480xRGBx272	400xRGBx240	
NCLK Frequency		FNCLK	33.2	9	8.3	MHz
Horizontal valid data		t_{hd}	800	480	400	NCLK
1 Horizontal Line		t_h	1056	525	528	NCLK
HSYNC Pulse Width	Min.	t_{hpw}	1			NCLK
	Typ.		-			
	Max.		-			
Hsync back porch		t_{hbp}	216	43	108	NCLK
Hsync front porch		t_{hfp}	40	2	20	NCLK
DEN Enable Time		t_{ep}	800	480	400	NCLK

Table 3.2 LCD horizontal timing parameters

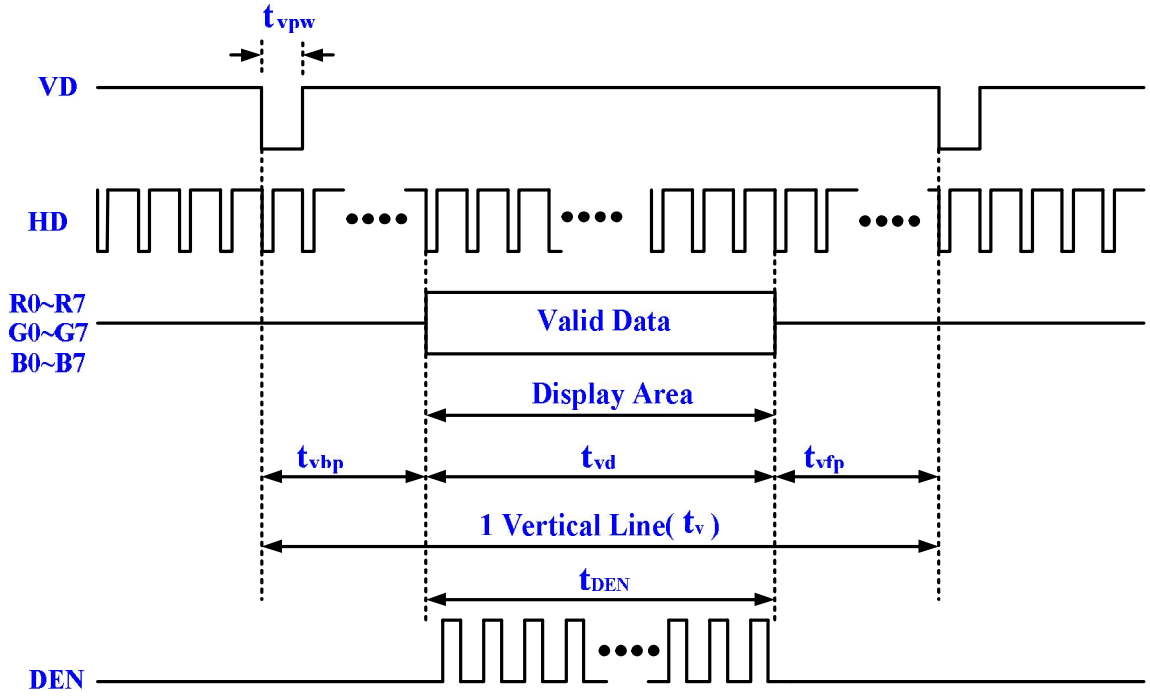


Figure 3.4 LCD vertical timing specification

Parameter		Symbol	Panel Resolution			Unit
			800xRGBx480	480xRGBx272	400xRGBx240	
Vertical valid data		t_{vd}	480	272	240	H
Vertical period		t_{v}	525	286	262	H
VSYNC Pulse Width	Min.	t_{vpw}	1			H
	Typ.		-			
	Max.		-			
Vertical back porch		t_{vbp}	35	12	20	H
Vertical front porch		t_{vfp}	10	2	2	H
Vertical blanking		t_{vb}	45	14	22	H
DEN Enable Time		T_{DEN}	480	272	240	H

Table 3.3 LCD vertical timing parameters

Parameter	Symbol	Min.	Unit
NCLK period	PW_{CLK*1}	25	ns
NCLK pulse high period	PWH_{*1}	10	ns
NCLK pulse low period	PWL_{*1}	10	ns
HD,VD, DEN, data setup time	t_{ds}	5	ns
HD,VD, DEN, data hold time	t_{dh}	5	ns

Table 3.4 The timing parameters of the LCD synchronous signals

3.3 The serial interface of the AD converter

This section will describe how to obtain the X/Y coordinates of the touch point from the AD converter.

The LTM also equipped with an Analog Devices AD7843 touch screen digitizer chip. The AD7843 is a 12-bit analog to digital converter (ADC) for digitizing x and y coordinates of touch points applied to the touch screen. The coordinates of the touch point stored in the AD7843 can be obtained by the serial port interface.

To obtain the coordinate from the ADC, the first thing users need to do is monitor the interrupt signal ADC_PENIRQ_n outputted from the ADC. By connecting a pull high resistor, the ADC_PENIRQ_n output remains high normally. When the touch screen connected to the ADC is touched via a pen or finger, the ADC_PENIRQ_n output goes low, initiating an interrupt to a FPGA that can then instruct a control word to be written to the ADC via the serial port interface.

The control word provided to the ADC via the DIN pin is shown in Table 3.5. This provides the conversion start, channel addressing, ADC conversion resolution, configuration, and power-down of the ADC. The detailed information on the order and description of these control bits can be found from the datasheet of the ADC in the *DATASHEET* folder on the LTM System CD-ROM.

MSB							LSB
S	A2	A1	A0	MODE	$SER/\overline{DEF}$	PD1	PD0

Bit	Mnemonic	Comment
7	S	Start Bit. The control word starts with the first high bit on DIN. A new control word can start every 15th DCLK cycle when in the 12-bit conversion mode, or every 11th DCLK cycle when in 8-bit conversion mode.
6-4	A2-A0	Channel Select Bits. These three address bits, along with the $SER/\overline{DEF}$ bit, control the setting of the multiplexer input, switches, and reference inputs.

3	MODE	12-Bit/8-Bit Conversion Select Bit. This bit controls the resolution of the following conversion. With 0 in this bit, the conversion has a 12-bit resolution, or with 1 in this bit, the conversion has a 8-bit resolution.
2	SER /$\overline{DEF}$	Single-Ended/Differential Reference Select Bit. Along with Bits A2 –A0, this bit controls the setting of the multiplexer input, switches, and reference inputs.
1,0	PD1,PD0	Power Management Bits. These two bits decode the power-down mode of the AD7843.

Table 3.5 Control register bit function description

Figure 3.5 shows the typical operation of the serial interface of the ADC. The serial clock provides the conversion clock and also controls the transfer of information to and from the ADC. One complete conversion can be achieved with 24 ADC_DCLK cycles. The detailed behavior of the serial port interface can be found in the datasheet of the ADC. Note that the clock (ADC_DCLK) and chip enable signals (SCEN) of the serial port interface SHRAE the same signal I/O with LCD driver IC. Users should avoid controlling the LCD driver IC and ADC at the same time when designing the serial port interface controller. Also, because the chip enable signal (SCEN) inputted to the ADC comes up with a logic inverter, the logic level of the SCEN should be inverse when it is used to control the ADC.

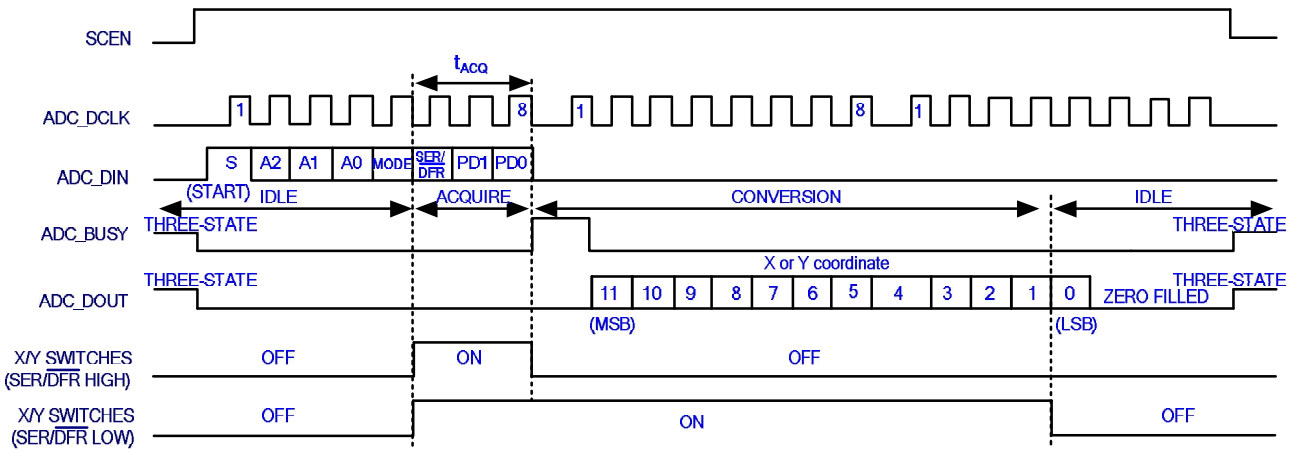


Figure 3.5 Conversion timing of the serial port interface

Chapter 4
Digital Panel Design Demonstration

This chapter illustrates how to exercise the LTM reference design provided with the kit. Users can follow the instructions in this chapter to build a 4.3 inch Ephoto demonstration and pattern generator using the DE2-115/DE2/DE1/DEN in 10 minutes.

4.1 Demonstration Setup

The demonstration configuration is illustrated as Figure 4.1. Display the bmp format photos, which are saved into the flash of DE2-115/DE2/DE1, on LTM module, through the control of FPGA on DE2-115/DE2/DE1 board. Users can change the displayed photo by touching the LTM touch panel.

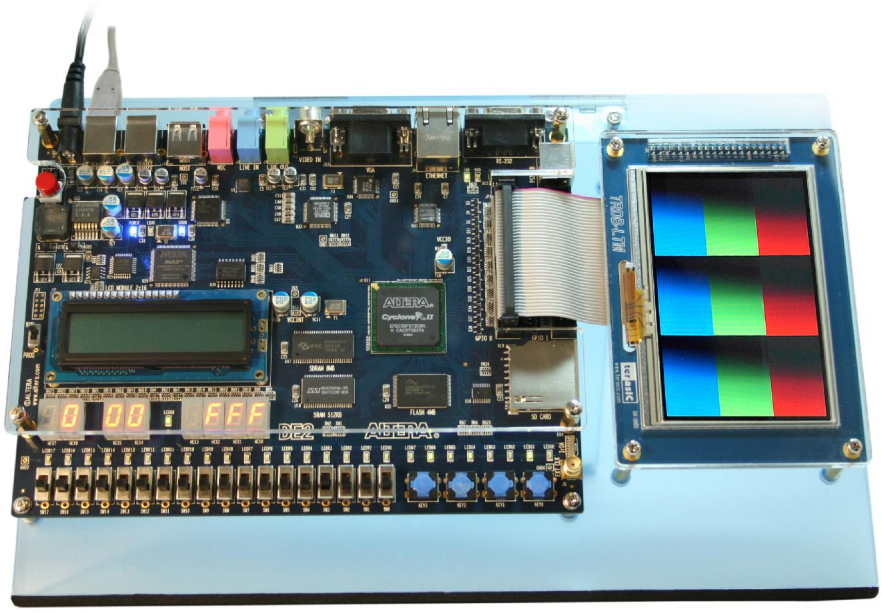


Figure 4.1. The Ephoto demonstration configuration setup

4.2 Loading Photos into the Flash

Locate the project directory from the CD-ROM and follow the steps below:

A: For Altera DE2-115 Board

Quartus II Project Directory: DE2_115_Control_Panel

FPGA Bitstream Used: DE2_115_ControlPanel.sof

B: For Altera DE2 Board

Quartus II Project Directory: DE2_Control_Panel_V1.04

FPGA Bitstream Used: DE2_USB_API.sof or DE2_USB_API.pof

C: For Altera DE1 Board

Quartus II Project Directory: DE1_Control_Panel_V1.00

FPGA Bitstream Used: DE1_USB_API.sof or DE1_USB_API.pof

1. Make sure the USB-Blaster download cable is connected into the host PC
2. Load the Control Panel bit stream (**DE2_USB_API/ DE1_USB_API**) into the FPGA. Please also refer to **Chapter 3 DE2/DE1 Control Panel** in the **Altera DE2/DE1 User Manual** for more details in the **Control Panel Software**

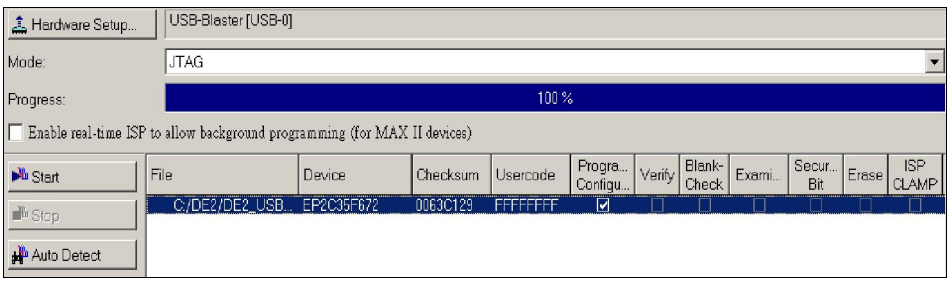


Figure 4.2. Programming window

3. Execute the Control Panel application software
4. Open the USB port by clicking Open > Open USB Port 0. The DE2/DE1 Control Panel application will list all the USB ports that connect to DE2/DE1 board
5. Switch to FLASH page and click on the "**Chip Erase(40 Sec)**" bottom to erase Flash data

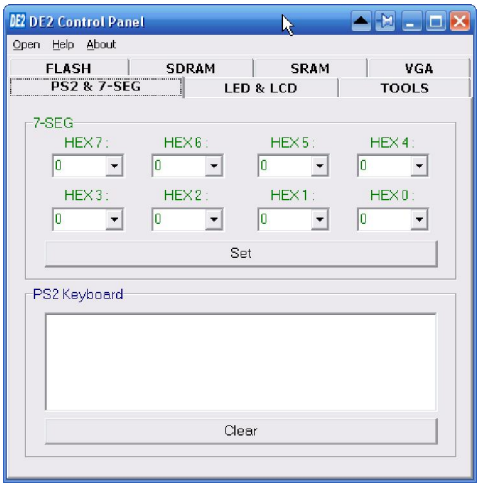


Figure 4.3. DE2 control panel window

6. Click on the "**File Length**" checkbox to indicate that you want to load the entire file
7. Click on the "**Write a File to FLASH**" bottom. When the Control Panel responds with the standard Windows dialog box and asks for the source file, select the "**480x3.bmp**" file in the "**Photo**" directory

4.3 Configuring the Ephoto Demonstration

Locate the project directory from the CD-ROM and follow the steps below:

A: For Altera DE2-115 Board

Quartus II Project Directory: DE2_115_LTM_Ephoto

FPGA Bitstream Used: DE2_115_LTM_Ephoto.sof or DE2_115_LTM_Ephoto.pof

B: For Altera DE2 Board

Quartus II Project Directory: DE2_LTM_Ephoto

FPGA Bitstream Used: DE2_LTM_Ephoto.sof or DE2_LTM_Ephoto.pof

C: For Altera DE1 Board

Quartus II Project Directory: DE1_LTM_Ephoto

FPGA Bitstream Used: DE1_LTM_Ephoto.sof or DE1_LTM_Ephoto.pof

1. Ensure the connection is made correctly as shown in Figure 4.4 and Figure 4.7. Make sure the IDE cable is connected to JP1 of the DE-115/DE2/DE1 board
2. Download the bitstream (DE2_115_LTM_Ephoto /DE2_LTM_Ephoto/ DE1_LTM_Ephoto) to the DE2/DE1 board
3. As shown in Figurate 4.5, touch the bottom left corner and top right corner of the LTM touch panel to display the next and previous photos respectively

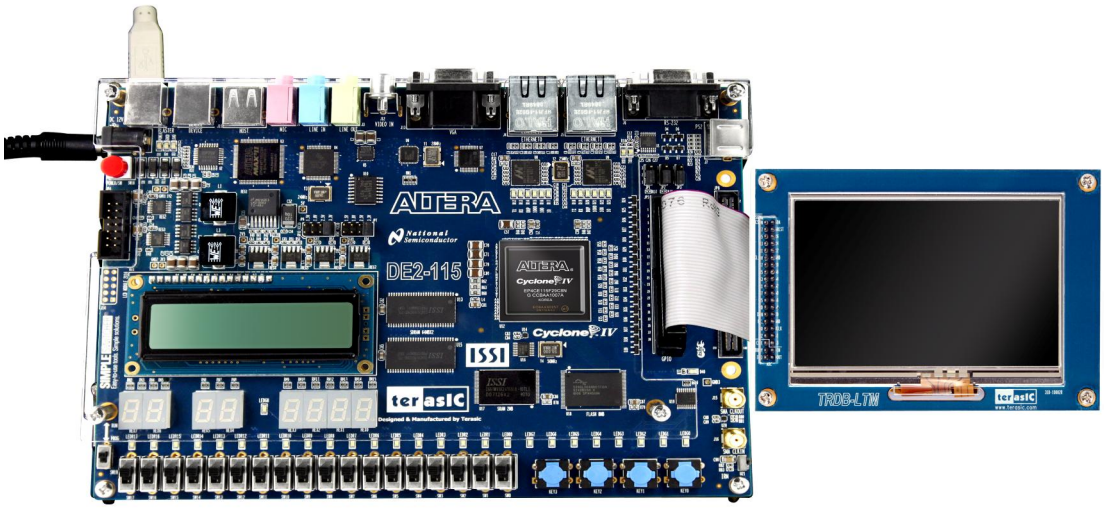


Figure 4.4. The connection setup for the Ephoto demonstration with DE2-115 board

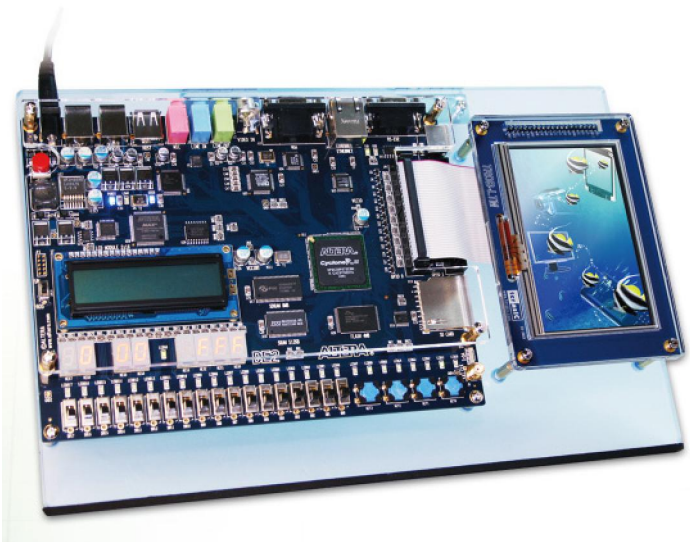


Figure 4.5. The connection setup for the Ephoto demonstration with DE2 board



Figure 4.6. The connection setup for the Ephoto demonstration with DE1 board

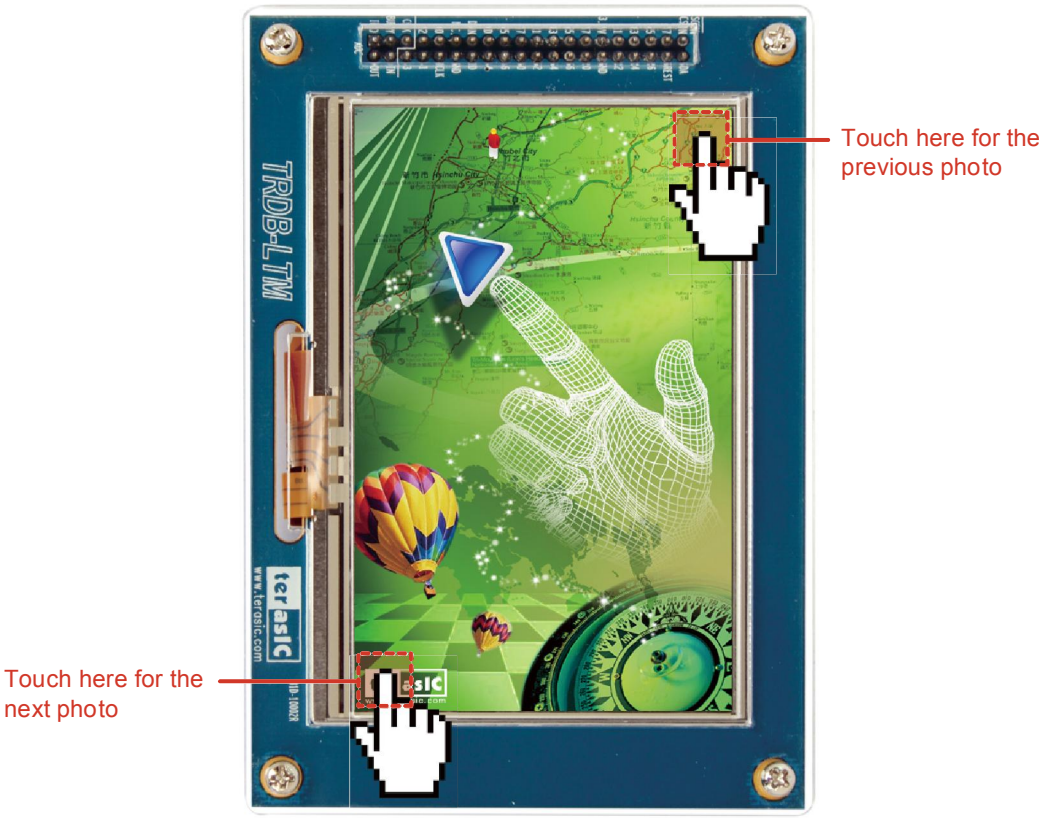


Figure 4.7. The touch function of changing displayed photo

4. When users touch the LTM screen, the 7-segment displays "HEX6~HEX4" and "HEX2~HEX0" on the DE2 board will display the X and Y coordinates (in Hexadecimal format) of the touch point respectively (DE2-115/DE2 board only). Figure 4.8 indicates that the x and y coordinates of the corners on the LTM touch screen

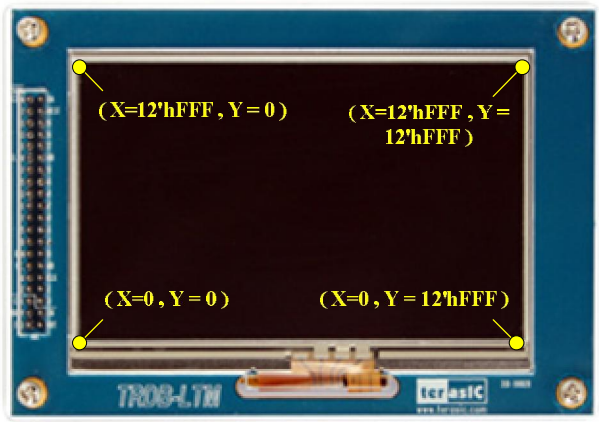


Figure 4.8. The x and y coordinates of the corners on the LTM touch screen

4.4 Block Diagram of the Ephoto Design

This section will describe the block diagram of the Ephoto demonstration to help users in reading the code provided.

Figure 4.9 shows the block diagram of the EPhoto demonstration. As soon as the bit stream is downloaded into the FPGA, the register values of the LCD driver IC using to control the LCD display function will be configured by the *LCD_SPI_Controller* block, which uses the serial port interface to communicate with the LCD driver IC. Meanwhile, the *Flash_to_SDRAM_Controller* block will read the RGB data of one picture stored in the Flash, and then write the data into SDRAM buffer. Accordingly, both the synchronous control signals and the picture data stored in the SDRAM buffer will be sent to the LTM via the *LCD_Timing_Controller* block.

When users touch LTM screens, the x and y coordinates of the touch point will be obtained by the *ADC_SPI_Controller* block through the ADC serial port interface. Then the *Touch_Point_Detector* block will determine whether these coordinates are in a specific range. If the coordinates fit the range, the *Touch_Point_Detector* block will control the *Flash_to_SDRAM_Controller* block to read the next or previous picture's data from the Flash and repeat the steps as mentioned before to command the LTM to display the next or previous picture.

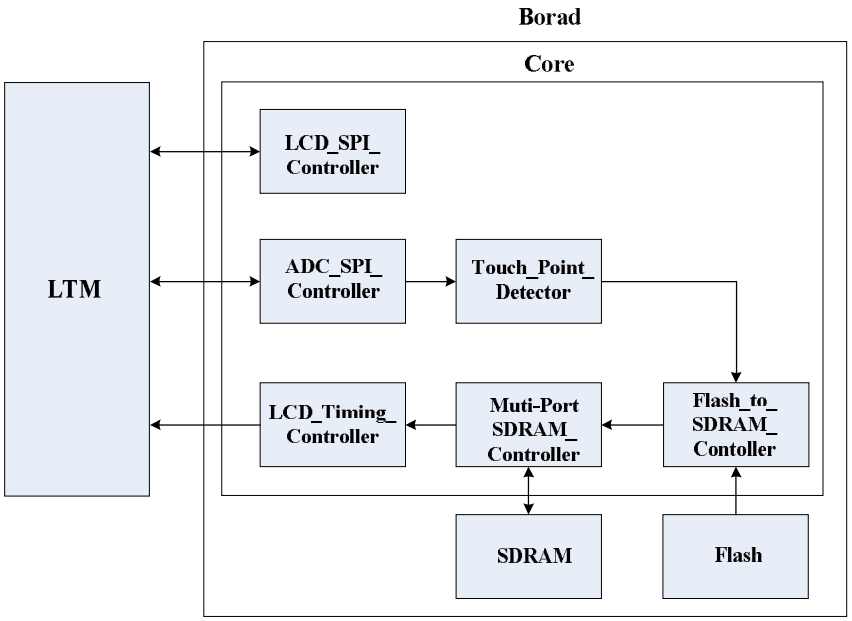


Figure 4.9. The block diagram of the Ephoto demonstration

4.5 Preprocessing the Desired Display Photo

If users want to display their own photos on the LTM ,they can follow the steps below:

1. Prepar three 24 bit *bmp* format photos and image resolution should be 800(high) x 480(width), as shown in Figure 4.10

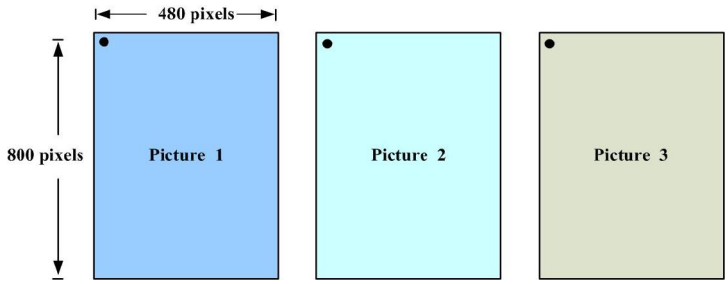


Figure 4.10. Original photo's resolution format

2. Use the image processing software (such as Photoshop or Photoimpact) to rotate the images counterclockwise and then merge these photos into a new photo image. The new photo's image resolution should be 1440(high) x 800 (width), as shown in Figure 4.11

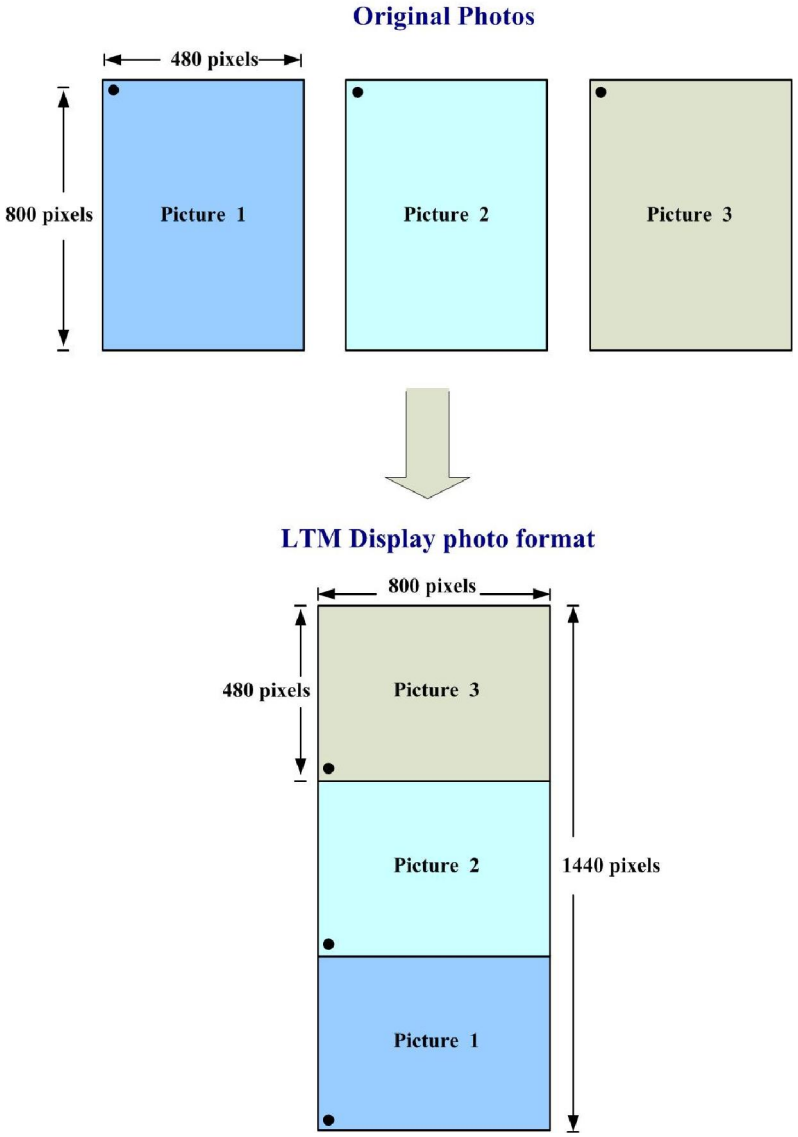


Figure 4.11 The photo format of the DE2_LTM_Ephoto/ DE1_LTM_Ephoto demonstration

4.6 Configuring the Pattern Generator for DE4/DE2-115/DE2/DE1 Board

Locate the project directory from the CD-ROM and follow the steps below:

A: For Altera DE4 Board

Quartus II Project Directory: DE4_LTM_Test

FPGA Bitstream Used: DE4_LTM_Test.sof or DE4_LTM_Test.pof

B: For Altera DE2-115 Board

Quartus II Project Directory: DE2_115_LTM_Test

FPGA Bitstream Used: DE2_115_LTM_Test.sof or DE2_115_LTM_Test.pof

C: For Altera DE2 Board

Quartus II Project Directory: DE2_LTM_Test

FPGA Bitstream Used: DE2_LTM_Test.sof or DE2_LTM_Test.pof

D: For Altera DE1 Board

Quartus II Project Directory: DE1_LTM_Test

FPGA Bitstream Used: DE1_LTM_Test.sof or DE1_LTM_Test.pof

1. Ensure the connection is made correctly as shown in Figure 4.12 and Figure 4.15 Make sure the IDE cable is connected to JP1 of the DE4/DE2-115/DE2/DE1 board

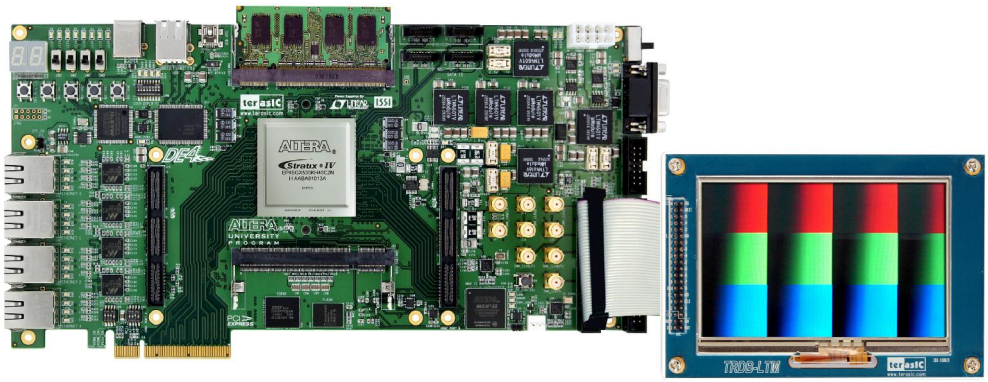


Figure 4.12. The connection setup for the pattern generator demo with DE4 board

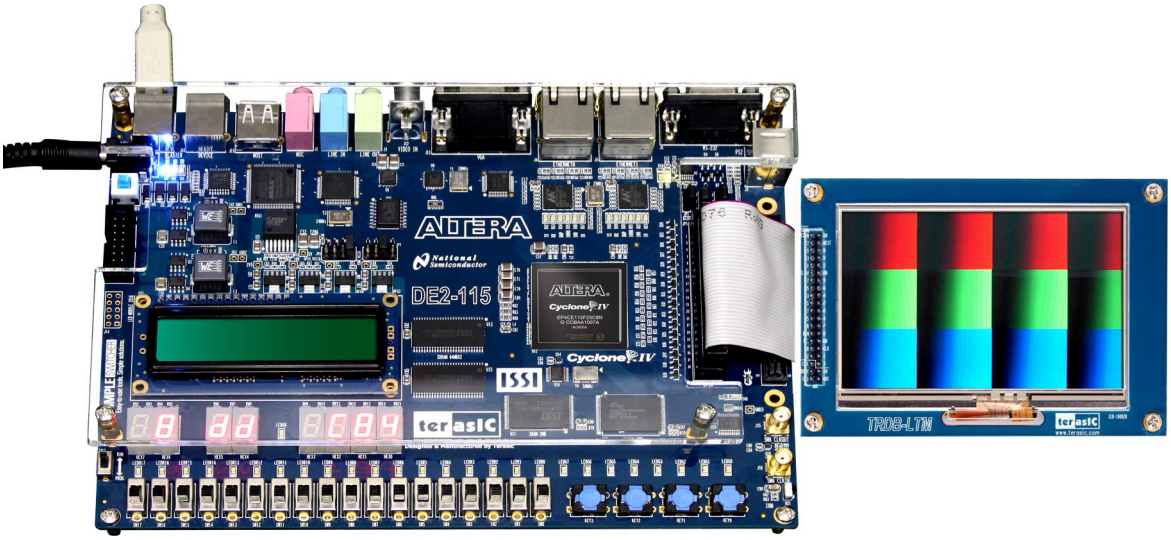


Figure 4.13. The connection setup for the pattern generator demo with DE2-115 board

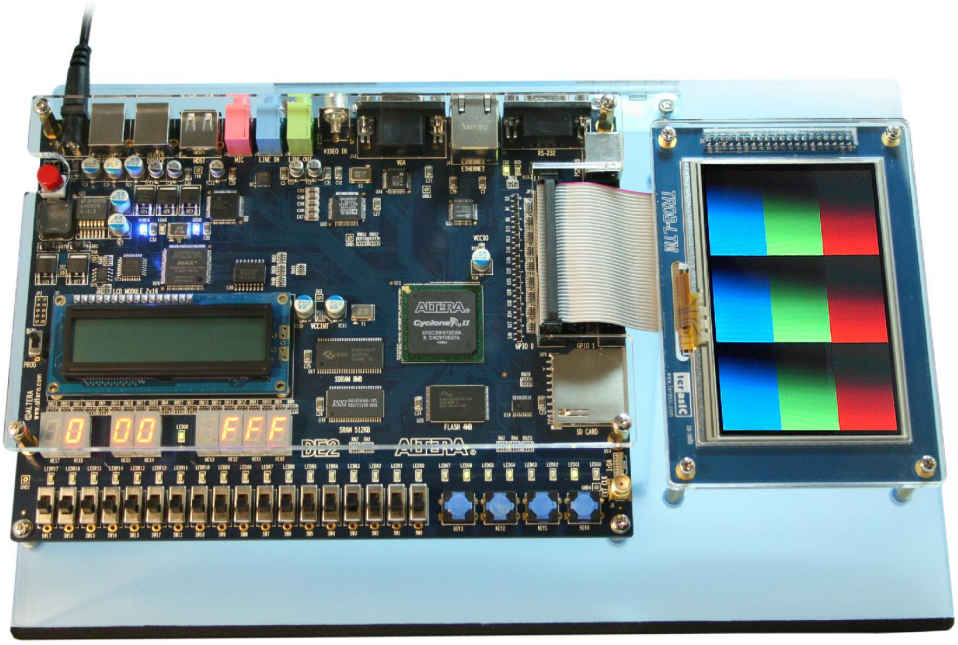


Figure 4.14. The connection setup for the pattern generator demo with DE2 board

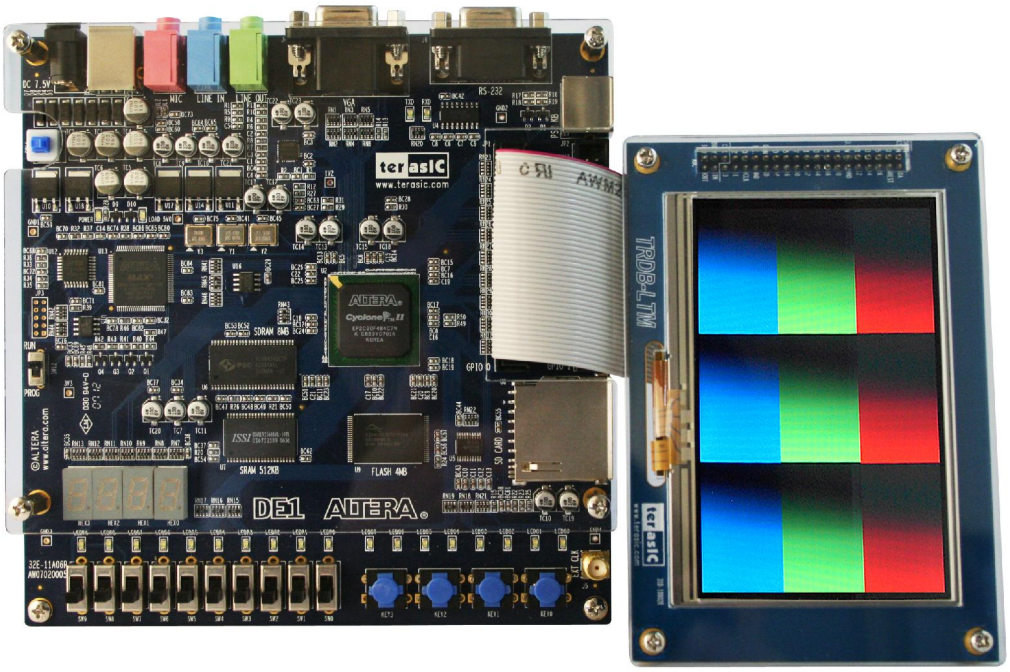


Figure 4.15. The connection setup for the pattern generator demo with DE1 board

2. Download the bitstream (DE2_115_LTM_Test/DE2_LTM_Test/ DE1_LTM_Test) to the DE2-115/DE2/DE1 board
3. Press KEY0 on the DE2-115/DE2/DE1 board to reset the circuit
4. Touch the LTM screen to switch to the other Pattern
5. The following table summarizes the Pattern type of this demonstration

Pattern
Gray bar
Color bar
50% gray level pattern
White pattern

6. When you touch the LTM panel, the 7-segment displays “HEX6~HEX4” and “HEX2~HEX0” on the DE2-115/DE2 will display the X and Y coordinates (in Hexadecimal format) of the LTM panel respectively (DE2-115/DE2 board only)

4.7 Configuring the Pattern Generator for DEN Board

Locate the project directory from the CD-ROM and follow the steps below:

Quartus II Project Directory:_DEN_LTM_Test

FPGA Bitstream Used:_DEN_LTM_Test.pof

1. Ensure the connection is made correctly as shown in Figure 4.16. Make sure the IDE cable is connected to the extension header of the DEN board

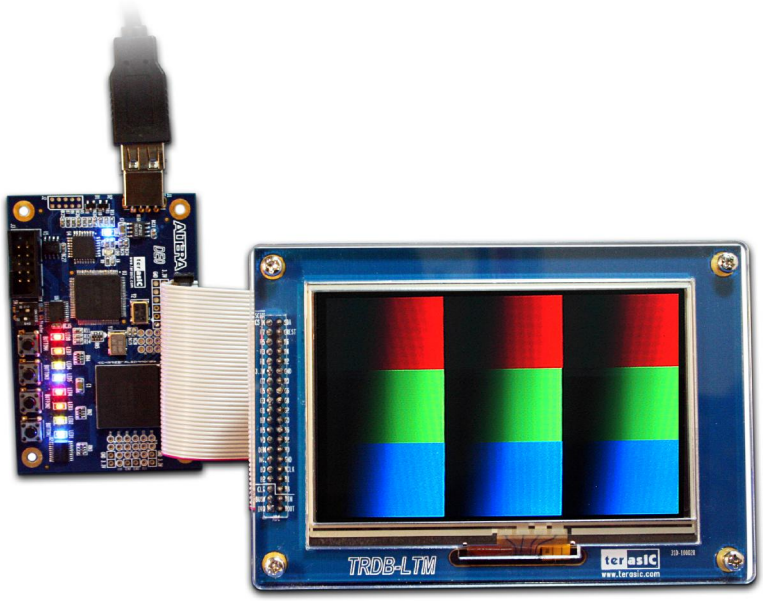


Figure 4.16. The connection setup for the pattern generator demo with DEN board

2. Download the bitstream (DEN_LTM_Test.pof) to the DEN board
3. Press BUTTON1 on the DEN board to reset the circuit
4. Touch the LTM screen to switch to the other Pattern
5. The following table summarizes the Pattern type of this demonstration

Pattern
Gray bar
Color bar
50% gray level pattern
White pattern

4.8 Picture Viewer Demonstration

This demonstration shows a simple picture viewer implementation using Nios II based SOPC system. It first read JPEG image from SD Card and then decode it using the Nios II. The frame buffers will be filled with decoded raw image data. The LTM will show the image that the buffer being displayed point to. When users touch the LTM, it will proceed to display the next buffered

image until there is no filled buffer. Figure 4.17 shows the block diagram of this demonstration.

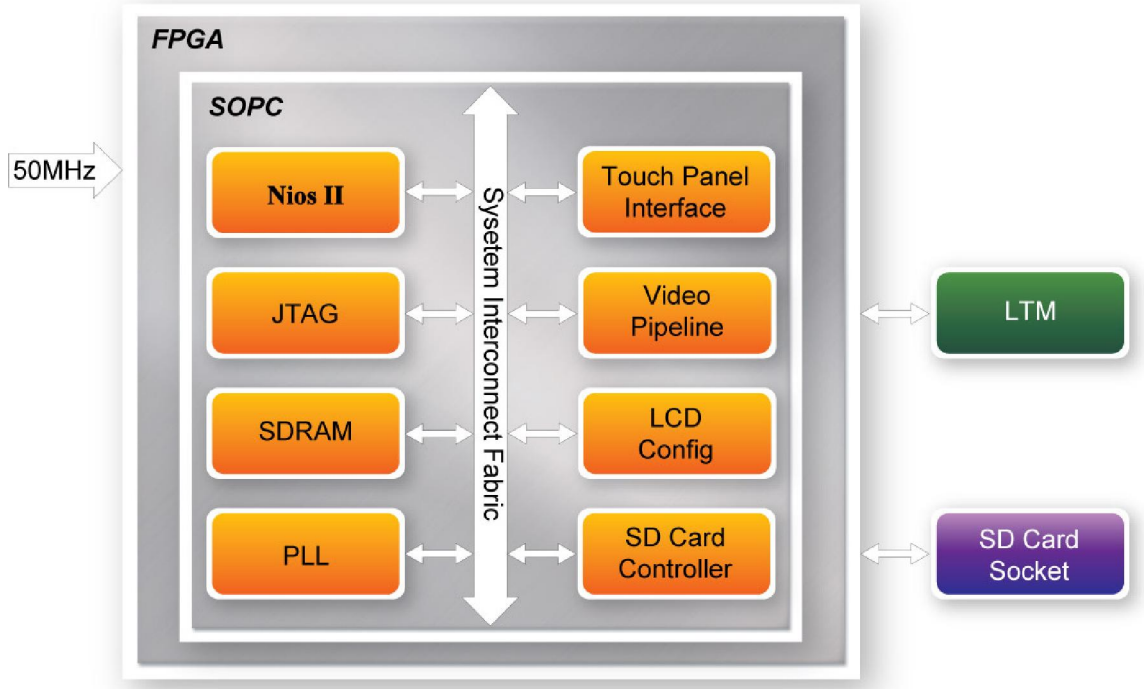


Figure 4.17. Block diagram of the picture viewer demonstration

Demonstration Setup, File Locations, and Instructions

- Project directory: DE2_115_LTM_Pic
- Bit stream used: DE2_115_LTM_Pic.sof
- Nios II Workspace: DE2_115_LTM_Pic\Software
- Format your SD Card into FAT16 format
- Place the jpg image files to the \jpg subdirectory of the SD Card. For best display result, the image should have a resolution of 800x480 or the multiple of that
- Connect the LTM to the DE2-115 board, insert the SD Card to the SD Card slot on the DE2-115, as shown in Figure 4.18
- Load the bitstream into the FPGA on the DE2-115 board
- Run the Nios II Software under the workspace DE2_115_LTM_Pic\Software
- Touch the play button will proceed to display the next image, as shown in Figure 4.19
- When there is no filled up frame buffers for display, it will enter into **Loading** mode. The top left corner will print information about the loading progress. In the **Loading** mode, program won't response to users' touch action. Table4.1 shows the instructions for running the demonstration

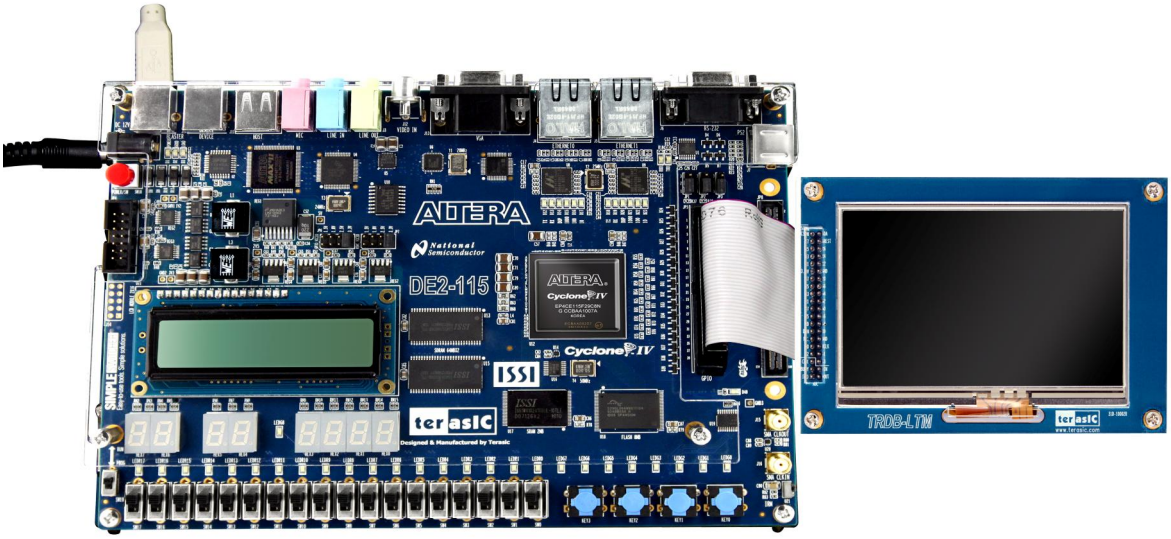


Figure 4.18 Picture viewer demonstration setup

Table 4.1 Touch panel displayed information

Display information	Implication
	Press the play button to display the next buffered image
	Indicates the loading progress

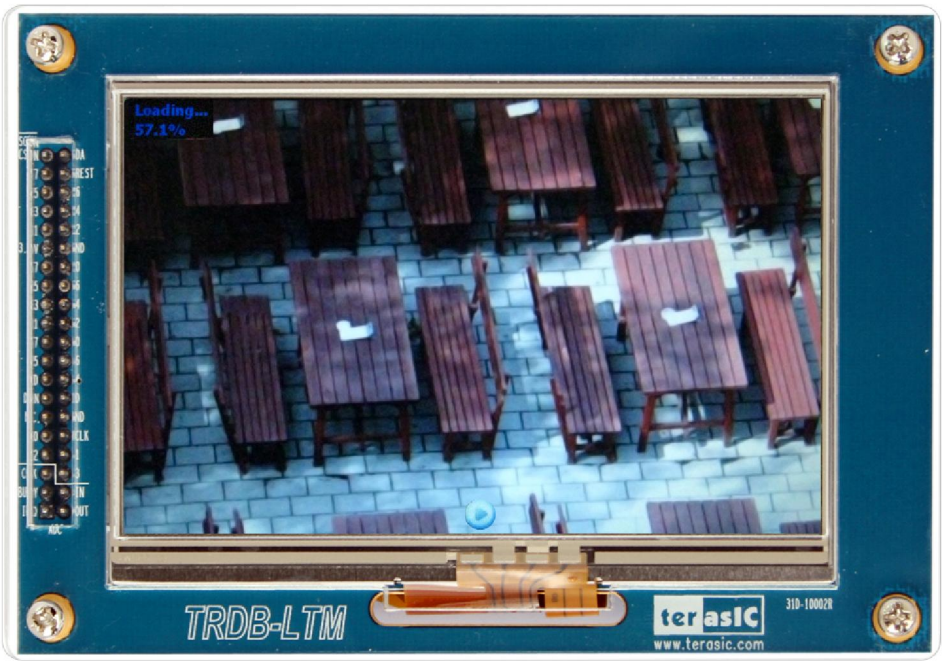


Figure 4.19 Picture viewer demonstration runtime screen shot

4.9 LTM Starter Demonstration

The LTM starter demonstration takes user the initial experience of an embedded system integrating a LCD Touch Panel. This demonstration consists of four sub item, Picture, Touch, Paint and Color pattern generator. The Picture program reads BMP images from SD Card and displays them on the LCD Touch Panel. As a response of your tap on the Touch Screen, it will display the next BMP image. The Touch segment draws a circle on where you touch the screen and updates its coordinates on the top left corner. The Paint segment is a little interesting. Your finger or touch pen slides on the Touch Panel. The software responds to every collected single touch and draws solid dots on the LCD Touch Panel. The formed curves represent your touch traces. There is also a color palette on the bottom side of the LCD, from where you could get your painting color. For best user-experience, you may need a touch pen. The pattern generator can be treated as an upgrade version of the LCD test program. The software successively generates different color patterns after a fixed time delay. Users could use it to quickly investigate any flaw of the LCD.

Figure 4.20 shows the hardware system block diagram of this demonstration. The system is clocked by an external 50MHz Oscillator. Through the internal PLL module, the generated 100MHz clock is used for Nios II processor and other components, and there also a 40MHz pixel clock for the video pipeline and 10MHz for low-speed peripherals. The Nios II CPU runs the application software and controls all the peripherals. A scatter-gather DMA is used to transfer pixel data from the video buffer to the video pipeline.

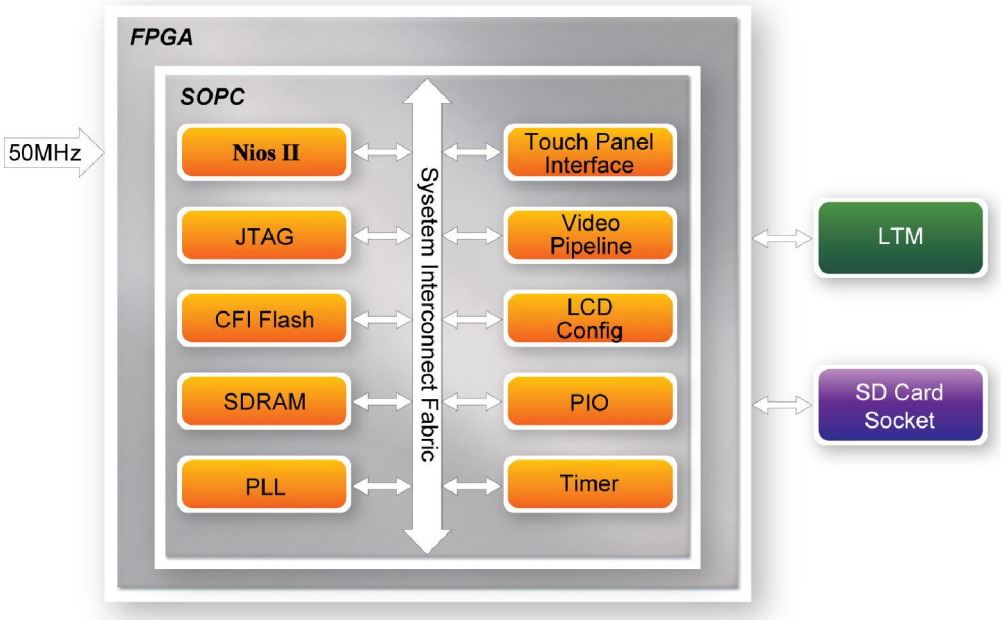


Figure 4.20 Block diagram of the LTM Starter demonstration

Figure 4.21 illustrates the software structure of this demonstration. The Nios II PIO block provides basic I/O functions to directly access the hardware. The SD Card block implements 4-bit mode protocol for communication with the SD Cards. The FAT File System block implements reading function for FAT16 and FAT32 file system. The touch panel SPI HAL block handles the bottom hardware responding and interface to upper layers. The sgdma HAL allocates required frame/descriptor buffers to specified memory address and is responsible of handling frame buffer update issue.

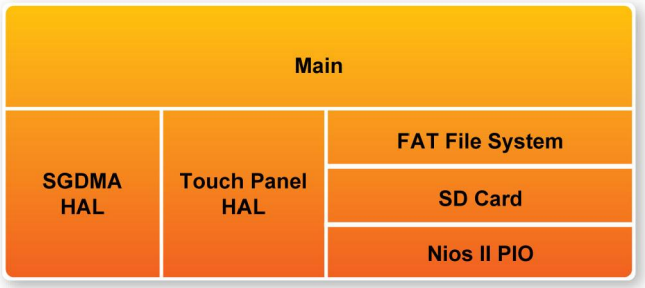


Figure 4.21. Software stack of the LTM Starter demonstration

Demonstration Source Code

- Project directory: DE2_115_LTM_Starter
- Bit stream used: DE2_115_LTM_Starter.sof
- Nios II Workspace: DE2_115_LTM _Starter\Software

Demonstration Batch File

Demo Batch File Folder: DE2_115_LTM _Starter \demo_batch

The demo batch file includes the following files:

- Batch File: DE2_115_LTM _Starter.bat, DE2_115_LTM _Starter_bashrc
- FPGA Configure File: DE2_115_LTM _Starter.sof
- Nios II Program: DE2_115_LTM _Starter.elf

Demonstration Setup

- Make sure Quartus II and Nios II are installed on your PC
- Power on the DE2-115 board
- Connect USB Blaster to the DE2-115 board and install USB Blaster driver if necessary
- Copy some bmp images to the root directory of the SD Card

- Insert the SD Card into the SD Card socket of LTM
- Execute the demo batch file "DE2_115_LTM _Starter.bat" under the batch file folder, DE2_115_LTM _Starter\demo_batch
- After Nios II program is downloaded and executed successfully, a prompt message will be displayed in nios2-terminal
- From on the touch panel, tap any icon of the main interface and start the experience
- Under each sub item, touch the **Exit** button on the left bottom corner will lead you back to the main interface. Figure4.22 and 4.23 give photos of sub items when running the DE2-115 LTM Starter demonstration



Figure 4.22 Touch sub item of the DE2-115 LTM Starter demonstration

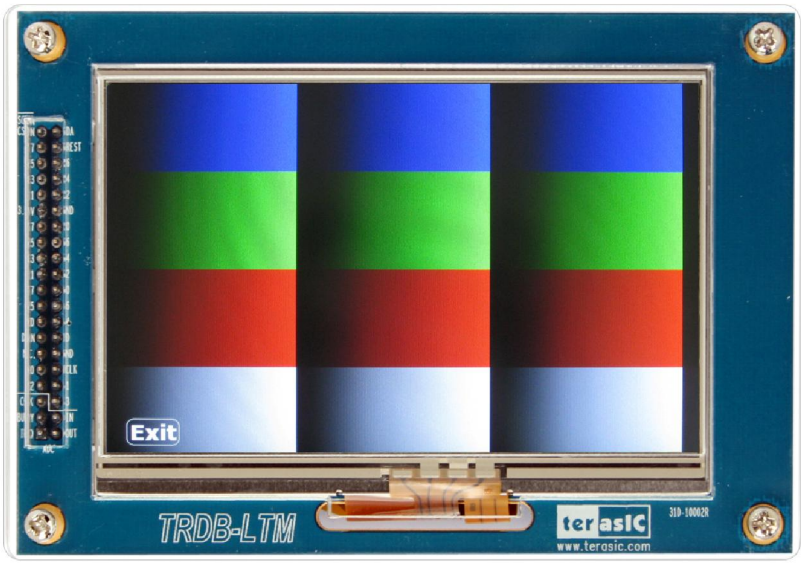


Figure 4.23 Color pattern sub item of DE2-115 LTM Starter demonstration

4.10 Video and Image Processing

The Video and Image Processing (VIP) Example Design demonstrates dynamic scaling and clipping of a standard definition video stream in either National Television System Committee (NTSC) or phase alternation line (PAL) format and picture-in-picture mixing with a background layer. The video stream is output in high definition resolution (800×480) on LTM

The example design demonstrates a framework for rapid development of video and image processing systems using the parameterizable MegaCore® functions that are available in the Video and Image Processing Suite. Available functions are listed in Table 4.2.

Table 4.2 VIP IP cores functions

IP MegaCore Function	*Description*
Frame Reader	Reads video from external memory and outputs it as a stream.
Control Synchronizer	Synchronizes the changes made to the video stream in real time between two functions.
Switch	Allows video streams to be switched in real time.
Color Space Converter	Converts image data between a variety of different color spaces such as RGB to YCrCb.
Chroma Resampler	Changes the sampling rate of the chroma data for image frames, for example from 4:2:2 to 4:4:4 or 4:2:2 to 4:2:0.
2D FIR Filter	Implements a 3 x 3, 5 x 5, or 7 x 7 finite impulse response (FIR) filter on an image data stream to smooth or sharpen images.

Alpha Blending Mixer	**Mixes and blends multiple image streams—useful for implementing text overlay and picture-in-picture mixing.**
Scaler	**A sophisticated polyphase scaler that allows custom scaling and real-time updates of both the image sizes and the scaling coefficients.**
Deinterlacer	**Converts interlaced video formats to progressive video format using a motion adaptive deinterlacing algorithm. Also supports 'bob' and "weave" algorithms**
Test Pattern Generator	**Generates a video stream that contains still color bars for use as a test pattern.**
Clipper	**Provides a way to clip video streams and can be configured at compile time or at run time.**
Color Plane Sequencer	**Changes how color plane samples are transmitted across the Avalon-ST interface. This function can be used to split and join video streams, giving control over the routing of color plane samples.**
Frame Buffer	**Buffers video frames into external RAM. This core supports double or triple-buffering with a range of options for frame dropping and repeating.**
2D Median Filter	**Provides a way to apply 3 x 3, 5 x 5, or 7 x 7 pixel median filters to video images.**
Gamma Corrector	**Allows video streams to be corrected for the physical properties of display devices.**
Clocked Video Input/Output	**These two cores convert the industry-standard clocked video format (BT-656) to Avalon-ST video and vice versa.**

These functions allow you to fully integrate common video functions with video interfaces, processors, and external memory controllers. The example design uses an Altera Cyclone® IV E EP4CE115F29 featured DE2-115 board.

A video source is input through an analog composite port on DE2-115 which generates a digital output in ITU BT656 format. A number of common video functions are performed on this input stream in the FPGA. These functions include clipping, chroma resampling, motion adaptive deinterlacing, color space conversion, picture-in-picture mixing, and polyphase scaling.

The input and output video interfaces on the DE2-115 are configured and initialized by software running on a Nios® II processor. Nios II software demonstrates how to control the clocked video input, clocked video output, and mixer functions at run-time is also provided. The video system is implemented using the SOPC Builder system level design tool. This abstracted design tool provides an easy path to system integration of the video processing data path with a NTSC or PAL video input, VGA output, Nios II processor for configuration and control. The Video and Image Processing Suite MegaCore functions have common open Avalon-ST data interfaces and Avalon Memory-Mapped (Avalon-MM) control interfaces to facilitate connection of a chain of video functions and video system modeling. In addition, video data is transmitted between the Video and Image Processing Suite functions using the Avalon-ST Video protocol, which facilitates building run-time controllable systems and error recovery.

Figure 4.24 shows the Video and Image Processing block diagram.

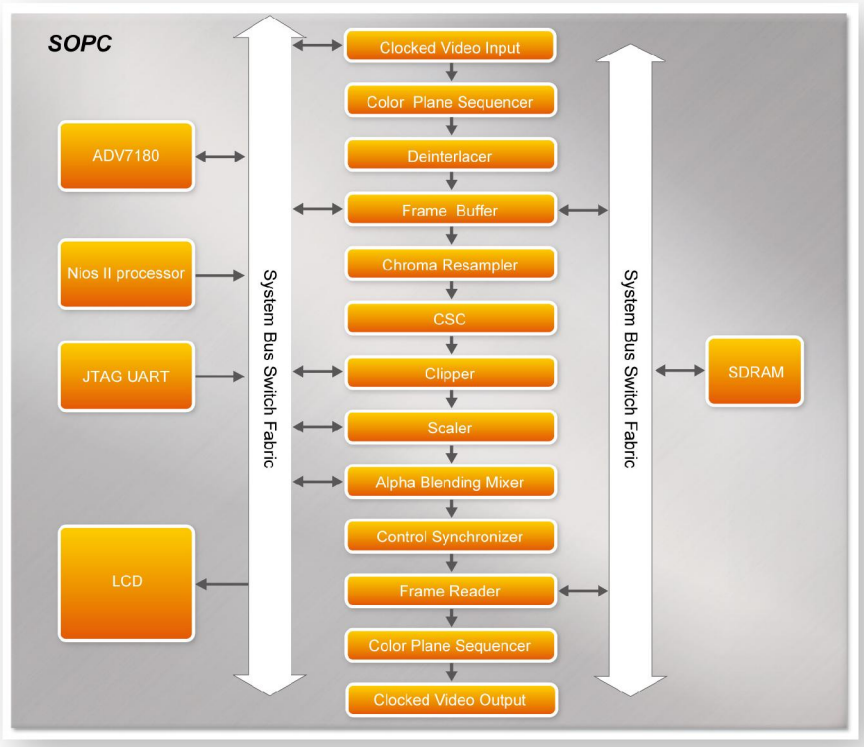


Figure 4.24 VIP Example SOPC Block Diagram (Key Components)

■ Demonstration Setup, File Locations, and Instructions

- Project directory: DE2_115_LTM_VIP
- Bit stream used: DE2_115_LTM_VIP.sof
- Connect a DVD player's composite video output(yellow plug) to the Video-IN RCA jack(J12) of the DE2-115 board. The DVD player has to be configured to provide NTSC output or PAL output
- Connect the VGA output of the DE2-115 board to a VGA monitor (both LCD and CRT type of monitors should work)
- Connect LTM to the GPIO port of DE2-115
- Load the bit stream into FPGA (note*)
- Run the Nios II and choose DE2_115_LTM_VIP \software as the workspace. Click on the Run button (note *)
- Press and drag the video frame box will result in scaling the playing window to any size. Figure 4.25 gives a photograph of the running result of the demonstration

Note: execute DE2_115_LTM_VIP \demo_batch\VIP.bat will download .sof and .elf files.



Figure 4.25 Running result of the DE2-115 LTM VIP demonstration

Chapter 5
Appendix

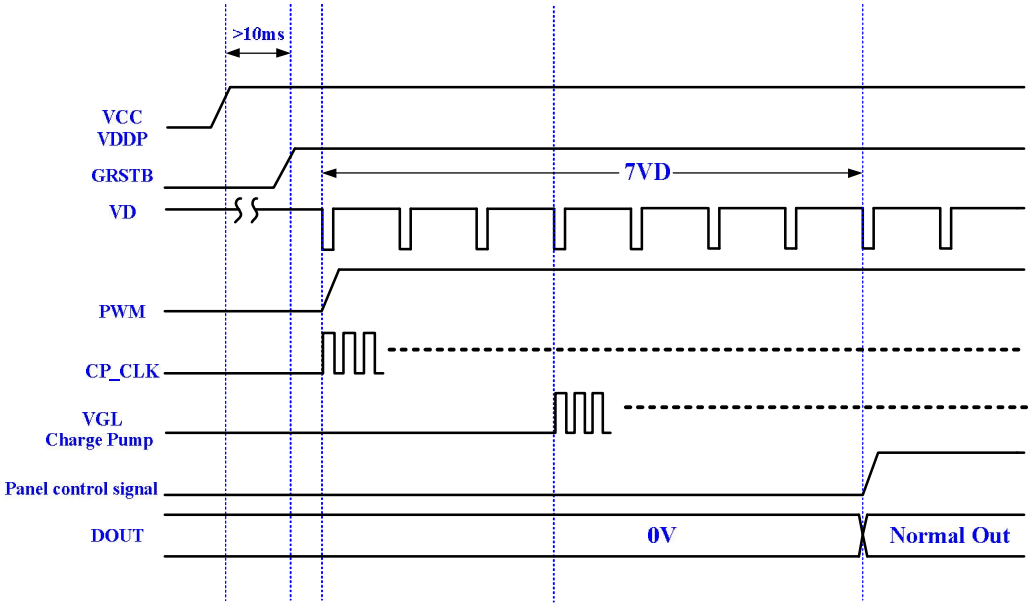
This chapter describes in more detail about the characteristic of the LCD for Hardware/Software developer's convenience.

5.1 Absolute Maximum Ratings of the LCD Panel Module

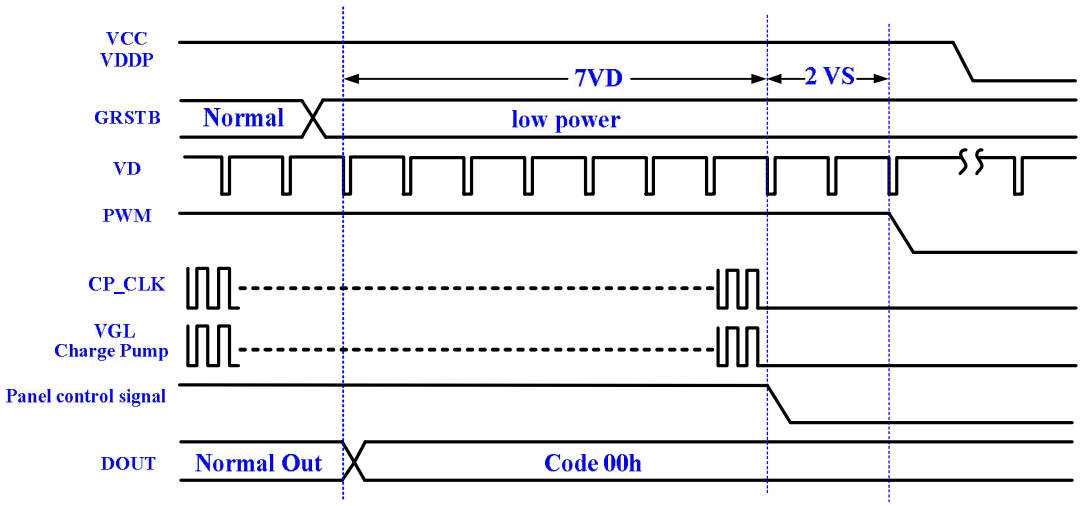
Item	Symbol	Min	Max	Unit	Remark
Logic Power Supply Voltage	V_{CC}	2.7	3.6	V	
Input Signal Voltage	V_{IN1}	0	V_{CC}	V	VD, HD, NCLK, SDA,SCL, SCEN,DEN,GREST
Back Light Forward Current	I_F	18	23	mA	
Operating Temperature	T_{OPR}	-20	+70	°C	
Storage Temperature	T_{STG}	-40	+85	°C	

5.2 Power ON/OFF and Mode Change Sequence of the LCD Panel Module

Power on (low power or reset mode to normal mode) sequence:



Power off (normal mode to low power mode) sequence :



5.3 Register map of the LCD Driver IC

Address	Default	Read/Write	Description
0x00	0x00	R/W	[7:0]:Testing register
0x01	0xC1	R	[7:4]:Chip ID [3:0]: Chip version
0x02	0x07	R/W	[7:6]: Dot inversion method selection [5]: VD polarity [4]: HD polarity [3]: Input clock latch data edge [2:0]: Resolution selection
0x03	0x5F	R/W	[7]: Hardware or software selection for resolution and standby. [6]:Pre-charge on/off [5:4]: Output driving capability [3]: PWM output on/off [2]: VGL pump output on/off

			[1]: CP_CLK output on/off[0]: Power management
0x04	0x17	R/W	[5:4]: VGL pump frequency [3:2]: CP_CLK frequency [1]: Vertical reverse mode [0]: Horizontal reverse mode
0x05	0x20	R/W	[5:0]: Horizontal start position for Sync mode
0x06	0x08	R/W	[3:0]: Vertical start position for Sync mode
0x07	0x20	R/W	[5:0]: CKH high pulse width
0x08	0x20	R/W	[5:0]: CKH non-overlap
0x09	0x20	R/W	[5:0]: ENB rising to CKH non-overlap
0x0A	0x20	R/W	[5:0]: ENB low pulse width
0x0B	0x20	R/W	[5:0]: R gain of contrast
0x0C	0x20	R/W	[5:0]: G gain of contrast
0x0D	0x20	R/W	[5:0]: B gain of contrast
0x0E	0x10	R/W	[5:0]: Offset of brightness R
0x0F	0x10	R/W	[5:0]: Offset of brightness G
0x10	0x10	R/W	[5:0]: Offset of brightness B
0x11	0x00	R/W	[7:6]: GAMMA 0[9:8] of gamma Correction [5:4]: GAMMA 8[9:8] of gamma Correction [3:2]: GAMMA 16[9:8] of gamma Correction [1:0]: GAMMA 32[9:8] of gamma Correction
0x12	0x5B	R/W	[7:6]: GAMMA 64[9:8] of gamma Correction [5:4]: GAMMA 96[9:8] of gamma Correction [3:2]: GAMMA 128[9:8] of gamma Correction [1:0]: GAMMA 192[9:8] of gamma Correction
0x13	0xFF	R/W	[7:6]: GAMMA 224[9:8] of gamma Correction [5:4]: GAMMA 240[9:8] of gamma Correction [3:2]: GAMMA 248[9:8] of gamma Correction [1:0]: GAMMA 256[9:8] of gamma Correction
0x14	0X00	R/W	[7:0]: GAMMA 0[7:0] of gamma Correction
0x15	0X20	R/W	[7:0]: GAMMA 8[7:0] of gamma Correction
0X16	0X40	R/W	[7:0]: GAMMA 16[7:0] of gamma Correction
0X17	0X80	R/W	[7:0]: GAMMA 32[7:0] of gamma Correction
0X18	0x00	R/W	[7:0]: GAMMA 64[7:0] of gamma Correction
0X19	0X80	R/W	[7:0]: GAMMA 96[7:0] of gamma Correction
0x1A	0x00	R/W	[7:0]: GAMMA 128[7:0] of gamma Correction
0x1B	0x00	R/W	[7:0]: GAMMA 192[7:0] of gamma Correction
0x1C	0X80	R/W	[7:0]: GAMMA 224[7:0] of gamma Correction
0x1D	0XC0	R/W	[7:0]: GAMMA 240[7:0] of gamma Correction
0x1E	0XE0	R/W	[7:0]: GAMMA 248[7:0] of gamma Correction
0x1F	0XFF	R/W	[7:0]: GAMMA 256[7:0] of gamma Correction
0x20	0xD2	R/W	[7:4]: Positive gamma output voltage level for source driver input FFH [3:0]: Positive gamma output voltage level for source driver input 00H
0X21	0xD2	R/W	[7:4]: Negative gamma output voltage level for source driver input FFH [3:0]: Negative gamma output voltage level for source driver input 00H
0X22	0x05	R/W	[3:0]: DC VCOM level

5.4 Register Definition of the LCD Driver IC

R02h:

R02[7:6]: Dot inversion method selection

R02[7:6]		Function	Note
0	0	Type1	Default (initial setting value)
0	1	Type2	
1	0	Type3	
1	1	Setting prohibited(Type1)	

R02[5]: VD polarity

R02[5]	Function	Note
0	Low pulse	Default (initial setting value)
1	High pulse	

R02[4]: HD polarity

R02[4]	Function	Note
0	Low pulse	Default (initial setting value)
1	High pulse	

R02[3]: Input clock latch data edge

R02[5]	Function	Note
0	Latch data at NCLK falling edge	Default (initial setting value)
1	Latch data at NCLK rising edge	

R02[2:0]: Resolution selection

R02[2:0]			Input Sequence	Output Resolution	Note
0	0	0	400RGBx240	800RGBx480	(Dual Scan)
0	0	1	480RGBx272	800RGBx480	(Dual Scan)
0	1	0	Setting Prohibited		
0	1	1	Setting Prohibited		
1	0	0	Setting Prohibited		
1	0	1	480RGBx272	480RGBx272	
1	1	0	Setting Prohibited		
1	1	1	800RGBx480	800RGBx480	Default

R03h:

R03[7]: Hardware or Software selection for resolution and standby

R03[7]	Function	Note
0	Hardware pin (RS[3:1,STBY])	Default (initial setting value)
1	Software register(R02[2:0,R03[0])	

R03[6]: Pre-charge ON/OFF

R03[6]	Function	Note
0	Pre-charge disable	
1	Pre-charge enable	Default (initial setting value)

R03[5:4]: Driving capability

R03[5:4]		Function	Note
0	0	75%	
0	1	100%	Default (initial setting value)
1	0	150%	
1	1	200%	

R03[3]: PWM output ON/OFF

R03[3]	Function	Note
0	PWM disable	
1	PWM enable	Default (initial setting value)

R03[2]: VGL pump output ON/OFF

R03[2]	Function	Note
0	VGL pump disable	
1	VGL pump enable	Default (initial setting value)

R03[1]: CP_CLK output ON/OFF

R03[1]	Function	Note
0	CP_CLK disable	
1	CP_CLK enable	Default (initial setting value)

Power management

R03[0]	Function	Note

0	Standby mode	
1	Normal operation	Default (initial setting value)

Note: In standby mode, DOUT [400:1], when VCOM are connected to GND, PWM disabled. Control signals STV, CKV, CKH1~CKH6, and XENB are low, while XCKV, XCKH1~XCKH6, and ENB are high. CP_CLK1 is pulled high and CP_CLK is pulled low

R04h:

R04[5:4]: VGL pump frequency

R04[5:4]		Period(Frequency for WVGA)	Note
0	0	2 * H(~16KHz)	
0	1	1 * H(~32KHz)	Default (initial setting value)
1	0	1/2 * H(~64KHz)	
1	1	1/4 * H(~128KHz)	

R04[3:2]: CP_CLK frequency

R04[5:4]		Period(Frequency for WVGA)	Note
0	0	2 * H(~16KHz)	
0	1	1 * H(~32KHz)	Default (initial setting value)
1	0	1/2 * H(~64KHz)	
1	1	1/4 * H(~128KHz)	

R04[1]: Vertical reverse function

R04[1]	Function	Note
0	Reverse (CSV=L)	
1	Normal (CSV=H)	Default (initial setting value)

R04[0]: Horizontal reverse function

R04[0]	Function	Note
0	Reverse	
1	Normal	Default (initial setting value)

R05h:

R05[5:0]: Horizontal display position shift for SYNC mode

R05[5:0]						Function	Note
0	0	0	0	0	0	-32 NCLK	

0	0	0	0	0	1	-31 NCLK	
:	:	:	:	:	:	:	(display shift right)
0	1	1	1	1	0	-2 NCLK	
0	1	1	1	1	1	-1 NCLK	
1	0	0	0	0	0	Center	Default (initial setting value)
1	0	0	0	0	1	+1 NCLK	
1	0	0	0	1	0	+2 NCLK	
:	:	:	:	:	:	:	(display shift left)
1	1	1	1	1	0	+30 NCLK	
1	1	1	1	1	1	+31 NCLK	

R06h:

R06[3:0]: Vertical display position shift for SYNC mode

R06[3:0]				Function	Note
0	0	0	0	-8 NCLK	
0	0	0	1	-7 NCLK	
:	:	:	:	:	(display shift down)
0	1	1	0	-2 NCLK	
0	1	1	1	-1 NCLK	
1	0	0	0	Center	Default (initial setting value)
1	0	0	1	+1 NCLK	
1	0	1	0	+2 NCLK	
:	:	:	:	:	(display shift up)
1	1	1	0	+6 NCLK	
1	1	1	1	+7 NCLK	

R07h~R0Ah:

R07[5:0]: CKH high pulse width adjustment. Set 0x20 for normal operation.

R08[5:0]: CKH non-overlap adjustment. Set 0x20 for normal operation (around 0.6us).

R09[5:0]: ENB to CKH1 non-overlap adjustment. Set 0x20 for normal operation (around 1.2us).

R0A[5:0]: ENB low pulse width adjustment. Set 0x20 for normal operation (around 2.8us).

Notice that CKV transition timing is in the middle of ENB low pulse.

R07~R0A[5:0]						Function	Note
0	0	0	0	0	0	-32 NCLK	
0	0	0	0	0	1	-31 NCLK	
:	:	:	:	:	:	:	
0	1	1	1	1	0	-2 NCLK	
0	1	1	1	1	1	-1 NCLK	
1	0	0	0	0	0	Center	Default (initial setting value)
1	0	0	0	0	1	+1 NCLK	
1	0	0	0	1	0	+2 NCLK	
:	:	:	:	:	:	:	
1	1	1	1	1	0	+30 NCLK	
1	1	1	1	1	1	+31 NCLK	

R0Bh:

R0B[5:0]: R Gain of Contrast

R0B[5:0]	R Gain of Contrast	Note
0x00	0.00000	
0x20	1.00000	Default (initial setting value)
0x3F	1.96875	

R0Ch:

R0C[5:0]: G Gain of Contrast

R0C[5:0]	G Gain of Contrast	Note
0x00	0.00000	
0x20	1.00000	Default (initial setting value)
0x3F	1.96875	

R0Eh:

R0E[5:0]: R Offset of Brightness

R0E[5:0]	R Offset of Brightness	Note
0x00	-16	
0x10	0	Default (initial setting value)
0x3F	47	

R0Fh:

R0F[5:0]: G Offset of Brightness

R0F[5:0]	G Offset of Brightness	Note
0x00	-16	
0x10	0	Default (initial setting value)
0x3F	47	

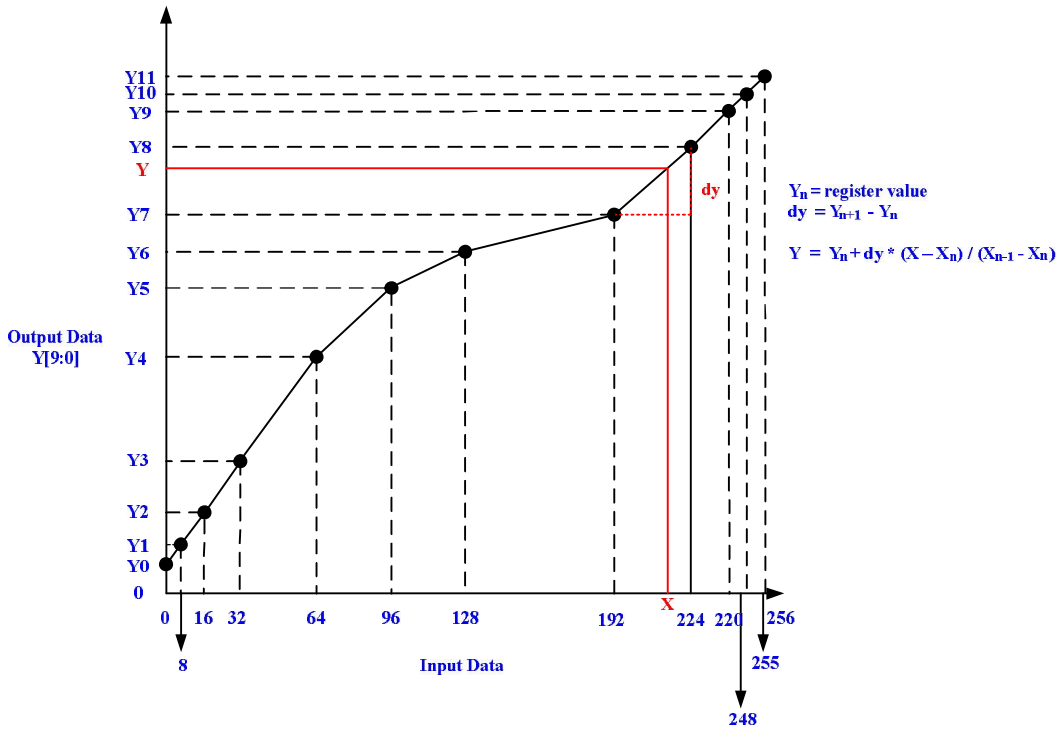
R10h:

R10[5:0]: B Offset of Brightness

R10[5:0]	B Offset of Brightness	Note
0x00	-16	
0x10	0	Default (initial setting value)
0x3F	47	

R11h ~ R1Fh: Gamma Correction

The gamma correction is done by 11-segment piecewise linear interpolation. The 11 segments are defined with 12 register values for level 0, 8, 16, 32, 64, 96,128, 192, 224, 240, 248, and 256 for positive polarity. Negative polarity data are generated symmetrically. The gamma correction output is then fed to 8-bit DAC and OP to drive the source lines on the panel.



R20h : Voltage range for positive polarity (when VDDP=5V and VDDN=-5V)

R20[7:4] defines positive polarity DAC reference voltage for code FFH

R20[7:4]	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xA	0xB	0xC	0xD (Default)	0xE	0xF
Gamma Level	3.3	3.4	3.5	3.6	3.7	3.8	3.9	4	4.1	4.2	4.3	4.4	4.5	4.6	4.7	4.8

R20[3:0] defines positive polarity DAC reference voltage for code 00H

R20[3:0]	0x0	0x1	0x2 (Default)	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
Gamma Level	0.2	0.25	0.3	0.35	0.4	0.45	0.5	0.55	0.6	0.65	0.7	0.75	0.8	0.85	0.9	0.95

R21 : Voltage range for negative polarity (when VDDP=5V and VDDN=-5V)

R21[7:4] defines negative polarity DAC reference voltage for code FFH

R21[7:4]	0x0	0x1	0x2	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xA	0xB	0xC	0xD (Default)	0xE	0xF
Gamma Level	-3.3	-3.4	-3.5	-3.6	-3.7	-3.8	-3.9	-4	-4.1	-4.2	-4.3	-4.3	-4.5	-4.6	-4.7	-4.8

R21[3:0] defines negative polarity DAC reference voltage for code 00H

R21[3:0]	0x0	0x1	0x2 (Default)	0x3	0x4	0x5	0x6	0x7	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
Gamma Level	-0.2	-0.25	-0.3	-0.35	-0.4	-0.45	-0.5	-0.55	-0.6	-0.65	-0.7	-0.75	-0.8	-0.85	-0.9	-0.95

R22 : DC VCOM level

R22[3:0] defines DC VCOM level

R22[3:0]	0x0	0x1	0x2	0x3	0x4	0x5 (Default)	0x6	0x7	0x8	0x9	0xA	0xB	0xC	0xD	0xE	0xF
VCOM Level	0.5	0.4	0.3	0.2	0.1	0	-0.1	-0.2	-0.3	-0.4	-0.5	-0.6	-0.7	-0.8	-0.9	-1

5.5 TV Revision History

Version	Change Log
V1.0	Initial Version (Preliminary)
V1.1	Edit appendix.
V1.2	Edit Ch3 and Ch4.
V1.2.1	Edit Figure 4.1, Figure 4.11, Figure 4.12, and Figure 4.13
V1.2.2	Edit chapter contents.
V1.2.3	Modify Table 2.1 and Figure 3.1
V1.2.4	Picture viewer, Starter, VIP section added DE2-115/DE4 related information added

5.6 Always Visit LTM Webpage for New Applications

We will continually provide interesting examples and labs on our LTM webpage. Please visit www.altera.com or **ltm.terasic.com** for more information.